BUET / PHYSICS 121

FORMULA ATLAS

THE DERIVATION & NUMERICAL EDITION

Waves & oscillations / Optics
Thermal physics



PREPARED FOR

EEE 25

TWO PARTS
Formulae / Symbols / Meaning

Part I navigator

Click a topic or use the PDF bookmarks. Source references use the printed pages of the question banks.

01	Simple harmonic motion	3	18	Resolution & polarization	10
02	Springs & oscillator energy	3	19	Wave plates & optical activity	11
03	Pendulums & two-body motion	3	20	Kinetic theory & molecular speeds	11
04	Superposition & Lissajous figures	4	21	Maxwell distributions	12
05	Damped oscillations	4	22	Equipartition & heat capacities	12
06	Forced oscillations & resonance	5	23	Real gases & critical constants	13
07	Electrical oscillations & radiation	6	24	First law & thermodynamic processes	13
08	Travelling waves	6	25	Adiabatic processes & steady flow	14
09	Standing waves & wave groups	7	26	Heat engines & Carnot cycle	14
11	Lenses & chromatic aberration	7	27	Refrigerators & entropy laws	15
13	Interference & double-source fringes	8	28	Entropy changes & phase transitions	15
14	Thin films, wedges & biprism	8	29	Thermodynamic potentials	16
15	Newton’s rings	9	30	Maxwell’s thermodynamic relations	17
16	Single-slit & circular diffraction	9	31	TdS equations & capacity difference	17
17	Double slit & diffraction grating	10	32	Phase equilibrium & Clapeyron equation	18

Source key

O: PHY121_Optics_Waves_Solutions(1).pdf
T: PHY121_Thermal_Physics_Solutions(1).pdf

Part I contains derivation results and supporting identities that are not used in the banks’ numerical solutions. Part II contains numerical-use formulae, including results shared with derivations. Classification follows the banks’ Part A / Part B sections; a calculation embedded in a theory question stays with Part I unless its formula also appears in Part B. Corrected or generalized variants follow the formula they support. Constants and conversions are collected at the end of Part II.

Simple harmonic motion

O: 3-10

Topic symbols. x, A: displacement/amplitude; t: time; f, T: frequency/period; v, a: velocity/acceleration

Oscillator equation of motion

$$m\ddot{x} + kx = 0$$

F: force (N); k: spring stiffness (N m^{-1}); x: displacement from equilibrium (m); U: potential energy (J); m: mass (kg); ω_0: natural angular frequency (rad s^{-1}). Dots denote time derivatives.
Use: connects restoring force to the oscillator equation; stiffness sets the restoring strength.

Springs & oscillator energy

O: 5-10

Topic symbols. x, A: displacement/amplitude; k: stiffness; v: velocity; E, K, U: total, kinetic, potential energy; $\omega_0 = 2\pi f$

Energy fractions, equal energies & cycle averages

$$\frac{U}{E} = \frac{x^2}{A^2}, \quad \frac{K}{E} = 1 - \frac{x^2}{A^2}, \quad K = U \implies x = \pm\frac{A}{\sqrt{2}}$$

$$\langle K \rangle = \langle U \rangle = \frac{E}{2}$$

$\langle\cdot\rangle$: average over one complete cycle. All other symbols defined in this topic's formula groups; ideal undamped SHM.

Effect of a uniform spring mass

$$m_{\text{eff}} = m + \frac{m_s}{3}, \quad T \simeq 2\pi\sqrt{\frac{m + m_s/3}{k}}$$

m_s: uniform spring mass; m: end load; m_{eff}: effective moving mass; T: period; h: static load extension; g: gravitational acceleration. The $m_s/3$ model assumes displacement varies linearly along the spring; it is an approximation.
Use: distinguishes the spring's inertial contribution from its static load extension.

Nonlinear spring compression

$$F = -k_1 x - k_2 x^3, \quad U = \frac{1}{2}k_1 x^2 + \frac{1}{4}k_2 x^4$$

$$\frac{1}{2}mv_i^2 = U(x_{\text{max}}), \quad x_{\text{max}} = \left[\frac{\sqrt{k_1^2 + 2k_2 m v_i^2} - k_1}{k_2}\right]^{1/2}$$

k_1: linear stiffness (N m^{-1}); k_2: cubic coefficient (N m^{-3}); v_i: incident block speed; x_{max}: maximum compression. Initially uncompressed; no dissipative loss; $k_1, k_2 > 0$.

Pendulums & two-body motion

O: 4-9

Physical pendulum: torque & equation

$$\tau = -Mgl\sin\theta \simeq -Mgl\theta, \quad I\ddot{\theta} + Mgl\theta = 0$$

$$I = I_G + Ml^2 = M(K_G^2 + l^2)$$

τ: restoring torque (N m); M: body mass; g: gravity; l: pivot-to-centre-of-mass distance; θ: small angular displacement; I: inertia about pivot (kg m^2); I_G: inertia about centre of mass; K_G: radius of gyration; T: period.

Conjugate pivots & minimum period

$$L_{\text{eq}} = l + \frac{K_G^2}{l}, \quad ll' = K_G^2, \quad T(l) = T(l')$$

$$l = K_G: \quad L_{\text{eq,min}} = 2K_G, \quad T_{\text{min}} = 2\pi\sqrt{\frac{2K_G}{g}}$$

L_{eq}: equivalent simple-pendulum length; l': centre-of-mass distance to conjugate pivot. Other symbols defined in this topic's formula groups; small oscillations.

Torsional pendulum

$$\tau = -\kappa\theta, \quad I\ddot{\theta} + \kappa\theta = 0, \quad \omega_0 = \sqrt{\frac{\kappa}{I}}, \quad T = 2\pi\sqrt{\frac{I}{\kappa}}$$

$$E = \frac{1}{2}I\dot{\theta}^2 + \frac{1}{2}\kappa\theta^2 = \frac{1}{2}\kappa\theta_{\text{max}}^2$$

κ: torsional stiffness (N m rad^{-1}); I: rotational inertia; θ_{max}: angular amplitude; E: energy. θ: twist angle; τ: torque; ω_0: angular frequency; T: period.

Relative coordinate & two-body equation

$$x = (x_1 - x_2) - \ell_0, \qquad \mu\ddot{x} + kx = 0$$

x_1, x_2: mass coordinates; ℓ_0: natural spring length; x: relative extension; m_1, m_2: masses; μ: reduced mass; k: stiffness; T: relative-motion period. Frictionless horizontal motion.
Meaning: reduced mass converts relative motion of two bodies into an equivalent one-body oscillator.

Superposition & Lissajous figures

O: 11-14; 38

Topic symbols. f_x, f_y: coordinate frequencies.

Collinear SHMs of the same frequency

$$x_1 = A\sin\omega t, \quad x_2 = B\sin(\omega t + \delta), \quad x = R\sin(\omega t + \epsilon)$$

$$R^2 = A^2 + B^2 + 2AB\cos\delta, \quad \epsilon = \operatorname{atan2}(B\sin\delta, A + B\cos\delta)$$

x_1, x_2: component displacements; x: resultant; A, B, R: amplitudes; ω: common angular frequency; t: time; δ: component phase difference; ϵ: resultant phase (rad).

Frequency ratio 1:2: general trajectory

$$x = a\sin\omega t, \quad y = b\sin(2\omega t + \phi)$$

$$\left[\frac{y}{b} - \left(1 - \frac{2x^2}{a^2}\right)\sin\phi\right]^2 = \frac{4x^2}{a^2}\left(1 - \frac{x^2}{a^2}\right)\cos^2\phi$$

ϕ: phase offset in the doubled-frequency motion; a, b: amplitudes; x, y: coordinates; ω: lower angular frequency. Physical trajectory: $|x| \le a$, $|y| \le b$.

Parabolic trajectory

$$\phi = \frac{\pi}{2}: \qquad \frac{y}{b} = 1 - \frac{2x^2}{a^2}$$

ϕ: phase offset in the doubled-frequency motion; a, b: amplitudes; x, y: coordinates; ω: lower angular frequency. Physical trajectory: $|x| \le a$, $|y| \le b$.

Damped oscillations

O: 15-19

Topic symbols. $\omega_0 = \sqrt{k/m}$

Damped equation & characteristic roots

$$m\ddot{x} + b\dot{x} + kx = 0, \qquad r_\pm = -\gamma_d \pm \sqrt{\gamma_d^2 - \omega_0^2}$$

m: mass; b: viscous damping constant (kg s^{-1}); k: stiffness; x: displacement; γ_d: amplitude decay rate (s^{-1}); p: source damping parameter; $r_\pm$: characteristic exponents.
Meaning: damping rate measures how rapidly motion decays; the characteristic roots determine whether oscillation is possible.

Underdamping: oscillation with a shrinking envelope

$$\gamma_d < \omega_0: \qquad x = A_0 e^{-\gamma_d t}\cos(\omega_d t + \phi)$$

ω_d: damped angular frequency; T_d: damped period; A_0: initial amplitude envelope; $A(t)$: envelope; ϕ: phase constant; t: time. $x(0) = A_0\cos\phi$, so initial displacement need not equal A_0.
Meaning: underdamping produces repeated crossings of equilibrium while the amplitude shrinks exponentially.

Critical damping: boundary case

$$\gamma_d = \omega_0, \quad b_c = 2\sqrt{mk}, \quad x = (C_1 + C_2 t)e^{-\gamma_d t}$$

b_c: critical damping constant; C_1 (m), C_2 (m s^{-1}): initial-condition constants. Critical damping has a repeated real root. For fixed mass and stiffness, it gives the fastest asymptotic decay among the nonoscillatory cases; it has no repeated oscillations.

Overdamping: two nonoscillatory decay modes

$$\gamma_d > \omega_0, \quad r_\pm = -\gamma_d \pm \sqrt{\gamma_d^2 - \omega_0^2}, \quad x = C_1 e^{r_+ t} + C_2 e^{r_- t}$$

$r_\pm$: distinct negative decay rates; C_1, C_2: displacement constants (m). The slow mode usually dominates late-time return. The body does not repeatedly oscillate, although whether it crosses equilibrium once can depend on its starting velocity.

Logarithmic decrement: a supporting decay identity

$$\Lambda = \frac{1}{N}\ln\frac{A_j}{A_{j+N}} = \gamma_d T_d$$

τ_A: amplitude e-folding time; N: number of periods; A_j, A_{j+N}: successive same-sign peak magnitudes separated by N periods; Λ: log decrement per period; $t_{1/2,A}$: amplitude half-life.

Meaning: amplitude mean lifetime is the time for the amplitude envelope to become 1/e (about 36.8 percent) of its starting value. A half-life refers to 50 percent, so it is different. Logarithmic decrement measures the amplitude loss per cycle.

Exact energy loss & mean dissipated power

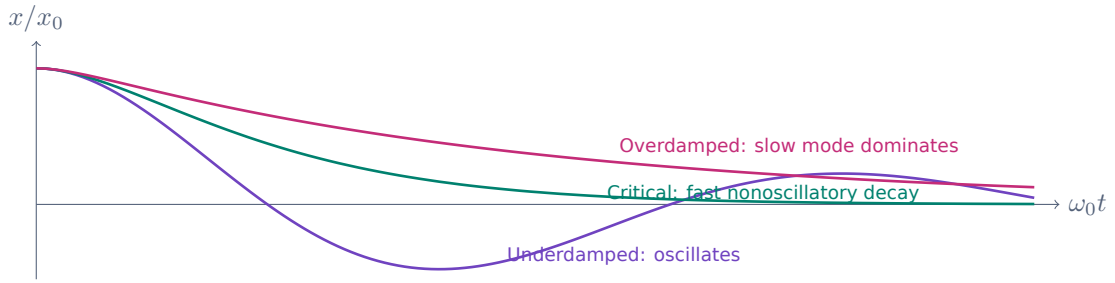
$$E_{\text{inst}} = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2, \quad -\frac{dE_{\text{inst}}}{dt} = b\dot{x}^2$$

$$\langle P_{\text{loss}}\rangle \simeq 2\gamma_d \overline{E}, \quad Q \simeq \frac{2\pi\overline{E}}{\Delta E_{\text{cycle}}}$$

E_{inst}: exact instantaneous mechanical energy; $\overline{E}$: slowly varying cycle-mean energy; E_0: initial mean energy; τ_E: energy e-folding time; Q: dimensionless quality factor; P_{loss}: dissipated power; ΔE_{cycle}: energy lost per cycle. Energy-envelope and cycle-loss expressions use weak damping.

Meaning: quality factor Q measures how weakly an oscillator loses energy relative to the energy it stores. Larger Q means slower relative loss and sharper resonance. Energy lifetime is the time for the energy envelope to fall to 1/e; it is half the amplitude lifetime.

Three damping cases: same initial displacement, zero initial velocity



x_0: initial displacement; horizontal axis is dimensionless time. Curves use $\gamma_d/\omega_0 = 0.23, 1, 2$, respectively.

Forced oscillations & resonance

O: 20-24

Topic symbols. $\omega_0 = \sqrt{k/m}$; $\gamma_d = b/(2m)$; $Q = m\omega_0/b$.

Driven equation & steady-state motion

$$m\ddot{x} + b\dot{x} + kx = F_0\cos\Omega t, \qquad x_{\text{ss}} = A(\Omega)\cos(\Omega t - \varphi)$$

m: mass; b: damping constant; k: stiffness; F_0: force amplitude (N); Ω: drive angular frequency; t: time; x_{ss}: steady displacement; A: its amplitude; $\omega_0 = \sqrt{k/m}$; $\gamma_d = b/(2m)$.

Use: predicts how strongly an oscillator responds after its initial transient has died away.

Phase lag & source sine convention

$$\varphi = \text{atan2}(b\Omega, k - m\Omega^2), \quad 0 \le \varphi \le \pi$$

$$\Omega \ll \omega_0 : \varphi \to 0; \quad \Omega = \omega_0 : \varphi = \frac{\pi}{2}; \quad \Omega \gg \omega_0 : \varphi \to \pi$$

$$x_{\text{ss}} = A\sin(\Omega t - \delta), \qquad \delta = \varphi - \frac{\pi}{2}$$

φ: displacement lag relative to a cosine driving force; δ: phase parameter if displacement is written as a sine. Other symbols defined in this topic's formula groups; $b > 0$, positive drive frequency.

Meaning: phase lag tells how far displacement falls behind the periodic driving force.

Exact displacement resonance: corrected peak

$$\Omega_r = \sqrt{\omega_0^2 - 2\gamma_d^2}, \quad \gamma_d < \frac{\omega_0}{\sqrt{2}}$$

$$A_{\text{max}} = \frac{F_0}{b\omega_d}$$

Ω_r: frequency of maximum displacement amplitude; ω_d: free damped angular frequency. For weaker damping, $\Omega_r \simeq \omega_0$ and $A_{\text{max}} \simeq F_0/(b\omega_0)$. If $\gamma_d \ge \omega_0/\sqrt{2}$, no nonzero-frequency displacement peak.

Use: locates the true maximum of displacement amplitude. It need not occur exactly at the natural frequency.

Mean absorbed power & sharpness

$$\langle P\rangle = \frac{1}{2}b\Omega^2A^2 = \frac{F_0^2 b\Omega^2}{2[(k-m\Omega^2)^2+b^2\Omega^2]}$$

$$\langle P\rangle_{\max} = \frac{F_0^2}{2b} \quad (\Omega=\omega_0), \quad \Delta\Omega_P = \frac{b}{m}, \quad Q = \frac{\omega_0}{\Delta\Omega_P}$$

$\langle P\rangle$: mean steady absorbed power (W); $\Delta\Omega_P$: separation of half-power frequencies. The displacement-squared bandwidth is approximately b/m only for weak damping.

Meaning: resonance bandwidth measures how wide the frequency range of strong energy absorption is. A narrow bandwidth means stronger frequency selectivity.

Electrical oscillations & radiation

O: 22-24

Series RLC equation

$$L\ddot{q} + R\dot{q} + \frac{q}{C} = \mathcal{E}_0 \sin\Omega t, \quad i = \dot{q}$$

$$m \leftrightarrow L, \quad b \leftrightarrow R, \quad k \leftrightarrow \frac{1}{C}, \quad F_0 \leftrightarrow \mathcal{E}_0$$

L: inductance (H); R: resistance (Ω); C: capacitance (F); q: capacitor charge (C); i: current (A); $\mathcal{E}_0$: emf amplitude (V); Ω: drive angular frequency; t: time. Mechanical analogues: m mass, b damping, k stiffness, F_0 force amplitude.

Impedance, current & phase

$$X_L = \Omega L, \quad X_C = \frac{1}{\Omega C}, \quad Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$I_0 = \frac{\mathcal{E}_0}{Z}, \quad i = I_0 \sin(\Omega t - \varphi_i), \quad \tan\varphi_i = \frac{X_L - X_C}{R}$$

X_L, X_C: inductive and capacitive reactances; Z: impedance magnitude (all ohms); I_0: current amplitude; φ_i: current lag angle. Positive φ_i: current lags; negative: current leads.

Resonant current

$$I_{0,\text{res}} = \frac{\mathcal{E}_0}{R}, \qquad Z_{\text{res}} = R, \quad \varphi_{i,\text{res}} = 0$$

ω_0, f_0: undamped resonance frequencies; ω_d, f_d: free damped discharge frequencies; Q: quality factor. Oscillatory discharge requires $R < 2\sqrt{L/C}$. At current resonance $Z = R$, $\varphi_i = 0$.

Meaning: RLC quality factor describes energy retention and resonance selectivity in the circuit.

Travelling waves

O: 24-30

Topic symbols. y, A: displacement/amplitude; x, t: position/time; f, ω: cyclic/angular frequency; $k_w = 2\pi/\lambda$: wavenumber; λ: wavelength; v_p: phase speed

Wave equation & local particle motion

$$\frac{\partial^2 y}{\partial t^2} = v_p^2 \frac{\partial^2 y}{\partial x^2}, \quad u = \frac{\partial y}{\partial t} = A\omega\cos(\omega t - k_w x + \phi)$$

$$a_y = \frac{\partial^2 y}{\partial t^2} = -\omega^2 y, \quad u = -v_p\frac{\partial y}{\partial x}, \quad u_{\max} = \omega A = \frac{2\pi A}{\lambda}v_p$$

u: particle velocity (m s^{-1}); a_y: particle acceleration; $u_{\max}$: maximum particle speed. The slope identity is for the displayed $+x$ travelling wave; v_p is not particle speed.

Mean kinetic & elastic energy densities

$$\langle e_K\rangle = \langle e_U\rangle = \frac{1}{4}\rho\omega^2A^2$$

e_K, e_U, e: kinetic, elastic and total energy per volume (J m^{-3}); ρ: medium density (kg m^{-3}); I: mean intensity (W m^{-2}); P: mean power; S: cross-sectional area. Linear harmonic mechanical plane wave; these are cycle averages.

Meaning: energy density is energy per volume; intensity is mean power crossing unit area.

Longitudinal elastic wave identities

$$B = \rho v_p^2, \quad p_{\text{ac}} = -B\frac{\partial y}{\partial x}, \quad p_0 = \rho v_p \omega A$$

$$I = \frac{p_0^2}{2\rho v_p}, \qquad \mathcal{I}(x,t) = p_{\text{ac}}(x,t)u(x,t)$$

B: bulk modulus (Pa); p_{ac}: acoustic excess pressure; p_0: its amplitude; $\mathcal{I}$: instantaneous intensity; u: local particle velocity. I: time-mean intensity.

Standing waves & wave groups

O: 26-31

Topic symbols. y: displacement; A: component amplitude; x, t: position/time; ω: angular frequency; $k_w = 2\pi/\lambda$: wavenumber; λ: wavelength; j: integer.

Standing-wave pressure & energy flow

$$y = 2A\cos k_w x \sin \omega t: \quad u = 2A\omega \cos k_w x \cos \omega t$$
$$p_{\text{ac}} = 2BAk_w \sin k_w x \sin \omega t, \quad \langle p_{\text{ac}} u \rangle = 0$$

B: bulk modulus; p_{ac}: excess pressure; u: particle velocity. Zero time-mean energy transfer; instantaneous energy flow can be nonzero.

Meaning: a pure standing wave has zero time-mean energy flux, even though instantaneous local flow can occur.

Unequal counterpropagating amplitudes

$$y = A_i \sin(\omega t - k_w x) + A_r \sin(\omega t + k_w x)$$
$$y = 2A_r \cos k_w x \sin \omega t + (A_i - A_r)\sin(\omega t - k_w x)$$
$$I_{\text{net}} = \frac{1}{2}\rho v_p \omega^2 (A_i^2 - A_r^2)$$

$A_i \geq A_r \geq 0$: incident and reflected amplitudes (same-phase reflection convention); ρ: density; v_p: phase speed; I_{net}: mean net energy flux. Do not infer net power from the residual travelling-wave amplitude alone.

Two close frequencies: carrier & envelope

$$y = 2A\cos\left(\frac{\Delta\omega\, t - \Delta k_w\, x}{2}\right)\sin(\bar{\omega}t - \bar{k}_w x)$$
$$v_g = \frac{d\omega}{dk_w} = v_p + k_w \frac{dv_p}{dk_w} = v_p - \lambda\frac{dv_p}{d\lambda}$$

$\Delta\omega = \omega_1 - \omega_2$, $\Delta k_w = k_1 - k_2$: component differences; $\bar{\omega} = (\omega_1 + \omega_2)/2$, $\bar{k}_w = (k_1 + k_2)/2$: means; v_g: group velocity. Non-dispersive: $dv_p/d\lambda = 0$, hence $v_g = v_p$.

Meaning: group velocity follows the envelope of nearby waves; phase velocity follows their individual crests.

Lenses & chromatic aberration

O: 33-37

Topic symbols. u, v: signed object/image distances

Dispersive power & axial chromatic aberration

$$\omega_c = \frac{n_v - n_r}{n - 1}, \quad f_r - f_v \simeq \omega_c f, \quad v_r - v_v \simeq \omega_c \frac{v^2}{f}$$
$$\Delta f \simeq -\frac{f\,\Delta n}{n - 1}$$

n_v, n_r: violet and red indices; n: reference (mean/yellow) index; ω_c: dispersive power; f_v, f_r: violet and red focal lengths; v_v, v_r: image distances; f, v: reference values. Small dispersion, paraxial optics; Δ denotes small signed change.

Meaning: dispersive power measures relative colour dispersion; chromatic aberration is the resulting separation of colour foci.

Circle of least chromatic confusion

$$d_{\text{cc}} \simeq D\frac{|v_r - v_v|}{2|v|} \simeq \frac{D\omega_c |v|}{2|f|}, \qquad |v| = |f|: \; d_{\text{cc}} \simeq \frac{1}{2} D\omega_c$$

d_{cc}: blur-circle diameter; D: lens aperture diameter; v: reference image distance; f: reference focal length. Small colour separation; positive converging-lens geometry used in the bank.

Best-form lens radius ratio

$$\frac{R_1}{R_2} = \frac{n(2n-1) - 4}{n(2n+1)}, \qquad n = 1.5: \quad \frac{R_1}{R_2} = -\frac{1}{6}$$

R_1, R_2: surface radii of the best-form thin lens for a parallel incident beam; n: refractive index. Second formula: source two-converging-lens arrangement with shared deviation; d: separation; f_1, f_2: focal lengths. This separation is setup-specific.

Abbe sine condition

$$n_o y_o \sin\theta_o = n_i y_i \sin\theta_i$$

n_o, n_i: object- and image-space indices; y_o, y_i: corresponding object/image heights; θ_o, θ_i: ray angles to the axis. R: curved-face radius; n: lens index in air; f: focal length. Sine condition accompanies correction of spherical aberration for an aplanatic system.

Interference & double-source fringes

O: 38-41; 44-45

Phase, amplitude & intensity

$$\delta = \frac{2\pi\Delta}{\lambda} + \delta_0, \quad R^2 = A^2 + B^2 + 2AB\cos\delta$$
$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\delta, \quad I_{\text{max,min}} = (\sqrt{I_1} \pm \sqrt{I_2})^2$$

Δ: optical path difference; λ: vacuum wavelength (with optical path in the same length units); δ_0: initial/reflection phase offset; δ: total phase difference; A, B, R: component/resultant amplitudes; I_1, I_2, I: intensities.

Equal beams & average over fringes

$$I_1 = I_2 = I_s: \quad I = 4I_s\cos^2\frac{\delta}{2}, \quad I_{\max} = 4I_s, \quad I_{\min} = 0$$
$$\langle I\rangle_\delta = I_1 + I_2$$

I_s: intensity of either beam alone; $\langle I\rangle_\delta$: average over a full phase interval 0 to 2π; other symbols defined in this topic's formula groups. Coherent beams with parallel polarizations.

Young fringes: path & position equations

$$\Delta = d\sin\theta \simeq \frac{dx}{L}, \quad x_m^{\text{bright}} = \frac{m\lambda L}{d}, \quad x_m^{\text{dark}} = \frac{(m+1/2)\lambda L}{d}$$
$$I(x) = 4I_s\cos^2\left(\frac{\pi dx}{\lambda L}\right)$$

d: source/slit separation; L: source-to-screen distance; x: signed screen coordinate from centre; θ: observation angle; m: integer fringe order; β: fringe width. Small angles, $L \gg d, |x|$; no extra phase offset.

Lloyd mirror

$$d = 2h, \quad \beta = \frac{\lambda L}{2h}, \quad \delta_0 = \pi$$
$$\Delta_{\text{bright}} = (m+1/2)\lambda, \qquad \Delta_{\text{dark}} = m\lambda$$

h: source height above mirror plane; d: source-to-virtual-image separation; L: effective screen distance; Δ: geometrical path difference in air; m: integer. One phase reversal on reflection; zero-path-difference fringe is dark.

Thin films, wedges & biprism

O: 40-45

Topic symbols. m: integer order; β: fringe width

Thin film: refraction & optical path difference

$$n_0\sin i = n\sin r, \quad \Delta = 2nt\cos r, \quad \Delta_{\text{eff}} = 2nt\cos r + \frac{\lambda}{2}$$

n_0: incident-medium index; n: film index; i, r: incidence/refraction angles; t: film thickness; λ: vacuum wavelength; $m = 0, 1, 2, \ldots$; Δ: propagation optical path difference. These reflection conditions require exactly one phase reversal; transmitted conditions interchange bright and dark.

Biprism: conjugate-lens separation measurement

$$d = \sqrt{d_1 d_2}, \qquad \frac{d_1}{d} = \frac{v}{u}, \quad \frac{d_2}{d} = \frac{u}{v}$$

a: slit-to-biprism distance; b: biprism-to-observation distance; L: total source-to-screen distance; x_0, x_N: micrometer positions N fringe intervals apart; d: virtual-source separation; d_1, d_2: image separations at conjugate lens positions; u, v: positive object/image distance magnitudes.

Newton's rings

O: 42-46

Topic symbols. n: film refractive index; λ: wavelength; m: ring order; p: positive order difference

Film thickness geometry

$$r^2 = 2Rt - t^2 \simeq 2Rt, \qquad t \simeq \frac{r^2}{2R}, \qquad D = 2r$$

R: curvature radius of lens lower surface; r: ring radius; D: ring diameter; t: local film thickness. Thin gap: $t \ll R$; normal incidence.

Squared ring diameter versus order: slope

$$\frac{d(D_m^2)}{dm} = \frac{4\lambda R}{n}$$

n_{liq}: liquid index (air index approximated as 1); $D_{m,\text{air}}, D_{m,\text{liq}}$: same-order diameters before/after filling. Absolute order formula assumes perfect central contact; a fractional dark order signals inconsistent data or a central gap.

Single-slit & circular diffraction

O: 43; 47-48; 51-54

Single-slit amplitude & intensity

$$\alpha = \frac{\pi a \sin\theta}{\lambda}, \quad E(\theta) = E_0 \frac{\sin\alpha}{\alpha}, \quad I(\theta) = I_0 \left(\frac{\sin\alpha}{\alpha}\right)^2$$

a: slit width; θ: observation angle; λ: wavelength in surrounding air; E_0, I_0: on-axis field amplitude and intensity; E, I: off-axis values; α: half the total phase span. Fraunhofer diffraction.

Meaning: the sinc-squared envelope gives the angular brightness pattern of a single slit.

Exact single-slit secondary maxima

$$\tan\alpha = \alpha \quad (\alpha \neq 0), \qquad \frac{I_j}{I_0} = \left(\frac{\sin\alpha_j}{\alpha_j}\right)^2$$

$$\alpha_1 = 4.4934\ldots, \qquad \alpha_2 = 7.7253\ldots$$

j: fringe order; α_j: exact nonzero root for a secondary maximum, or its midpoint approximation as marked. Exact first two positive roots: 4.4934, 7.7253; corresponding intensity ratios 0.0472, 0.0165.

Approximate secondary-peak intensities

$$\frac{I_j}{I_0} \simeq \frac{4}{(2j+1)^2\pi^2}$$

j: fringe order; α_j: exact nonzero root for a secondary maximum, or its midpoint approximation as marked. Exact first two positive roots: 4.4934, 7.7253; corresponding intensity ratios 0.0472, 0.0165.

Circular aperture: exact Airy intensity

$$z = \frac{\pi D \sin\theta}{\lambda}, \quad I(\theta) = I_0 \left[\frac{2J_1(z)}{z}\right]^2$$

D: circular aperture diameter; J_1: first-order Bessel function; z: dimensionless argument; θ_1: first dark-ring angle; F: focal length; r_A, D_A: Airy-disc radius and diameter. The bank also uses $F\lambda/D$ as a slit-like estimate; it omits the circular-aperture factor 1.22.

Meaning: the Airy disc sets the circular-aperture diffraction blur; its first-dark-ring factor is 1.22.

Double slit & diffraction grating

O: 48-53

Topic symbols. $d = a + b$: pitch

Two finite slits

$$d = a + b, \quad \alpha = \frac{\pi a \sin\theta}{\lambda}, \quad \beta = \frac{\pi d \sin\theta}{\lambda}$$

$$I = 4I_s \left(\frac{\sin\alpha}{\alpha}\right)^2 \cos^2\beta$$

a: open-slit width; b: opaque gap width; d: centre-to-centre pitch; θ: diffraction angle; λ: wavelength; I_s: one slit on-axis intensity; α, β: phase parameters. b is not centre separation.

N-slit grating intensity & extrema

$$I = I_s \left(\frac{\sin\alpha}{\alpha}\right)^2 \left(\frac{\sin N\beta}{\sin\beta}\right)^2$$

$$I_{\text{principal}} = N^2 I_s \left(\frac{\sin\alpha}{\alpha}\right)^2$$

$$Nd\sin\theta = j\lambda, \quad j \notin N\mathbb{Z} \text{ (interference minima)}$$

$$\tan N\beta = N\tan\beta, \qquad \frac{I_{\text{sec}}}{I_{\text{peak}}} = \frac{1}{1 + (N^2 - 1)\sin^2\beta}$$

N: illuminated slit/line count; j: integer excluding multiples of N; $\mathbb{Z}$: integers; I_s, α, β, a, d defined in this topic's formula groups. Between principal peaks: $N-1$ interference minima and $N-2$ subsidiary peaks. $I_{\text{sec}}/I_{\text{peak}}$ compares the interference-factor secondary maximum to its principal peak. The tangent equation and ratio neglect variation of the slit envelope.

Grating angular dispersion

$$\frac{d\theta}{d\lambda} = \frac{m}{d\cos\theta}, \qquad \Delta\theta \simeq \frac{m\Delta\lambda}{d\cos\theta}$$

m: fixed spectral order; $\Delta\lambda = \lambda_2 - \lambda_1$: wavelength difference; $\Delta\theta$: angular separation (rad); d: pitch; θ: mean diffraction angle for the small-separation approximation.

Resolution & polarization

O: 50-58

Topic symbols. λ: wavelength; $\Delta\lambda$: wavelength separation.

Microscope resolution & numerical aperture

$$\text{NA} = n\sin\alpha \simeq \frac{nD}{2F}, \quad d_{\text{Abbe}} = \frac{\lambda}{2\text{NA}}, \quad d_{\text{Rayleigh}} = \frac{0.61\lambda}{\text{NA}}$$

n: object-space index; α: objective half-aperture angle; D, F: objective diameter/focal length; NA: numerical aperture; d: resolution distance. The bank uses the Abbe form; point-source Rayleigh uses 0.61. $\mathcal{R}_\lambda$: grating spectral resolving power; m: nonzero order; N: illuminated lines; $\lceil\cdot\rceil$: round up.

Meaning: resolving power quantifies the ability to distinguish nearby details or wavelengths; a smaller resolution limit is better.

Brewster, Snell & critical angle

$$\tan i_B = \frac{n_2}{n_1}, \quad n_1\sin i_B = n_2\sin r_B, \quad i_B + r_B = \frac{\pi}{2}$$

$$\sin\theta_c = \frac{n_2}{n_1} \quad (n_1 > n_2), \qquad v = \frac{c}{n}$$

n_1, n_2: incident/transmitted refractive indices; i_B: Brewster angle; r_B: refraction angle there; θ_c: total-internal-reflection critical angle; v: light phase speed; c: vacuum light speed; n: relevant index.

Wave plates & optical activity

O: 55-58

Topic symbols. θ: optical rotation; ℓ: solution length; c_s: concentration.

Birefringence & retardation

$$v_o = \frac{c}{n_o}, \quad v_e = \frac{c}{n_{e,\text{eff}}}, \quad \Delta n = |n_o - n_{e,\text{eff}}|$$

$$\Delta_{\text{opt}} = \Delta n\, t, \quad \delta = \frac{2\pi \Delta n\, t}{\lambda}$$

n_o: ordinary index; $n_{e,\text{eff}}$: direction-dependent extraordinary effective index; v_o, v_e: corresponding phase speeds; c: vacuum light speed; t: plate thickness; Δ_{opt}: path retardation; δ: phase-retardation magnitude; λ: vacuum wavelength. Use appropriate effective indices for the propagation direction.

Half-wave plate thickness & retardation

$$t_{\text{HWP}} = \frac{(2j+1)\lambda}{2\Delta n}, \quad \delta_{\text{HWP}} = (2j+1)\pi \qquad (j = 0, 1, \ldots)$$

j: plate order; t: thickness; Δn: birefringence magnitude; λ: wavelength. Minimum-thickness plates use $j = 0$. QWP at $45°$ to principal axes gives circular polarization; axis-aligned input remains linear.
Meaning: a wave plate delays one principal polarization component relative to the other, changing the polarization state.

Half-wave polarization rotation

$$\psi_{\text{out}} = 2\alpha - \psi_{\text{in}} \pmod{\pi}, \qquad |\Delta\psi| = 2|\theta|$$

α: fast-axis orientation; $\psi_{\text{in}}, \psi_{\text{out}}$: incident and emergent linear polarization orientations; $\theta = \psi_{\text{in}} - \alpha$: input-to-axis angle. Rotation magnitude understood modulo the equivalence of linear axes.

Polarimeter readings & concentration graph

$$\theta = R_{\text{sample}} - R_{\text{blank}}, \quad R = \text{MSR} + \text{VSR} \times \text{VC}$$

$$[\alpha]_\lambda^T = \frac{1}{\ell}\frac{d\theta}{dc_s}$$

R: angular reading; MSR: main-scale reading; VSR: coincident vernier division; VC: vernier constant (degrees per division); $R_{\text{sample}}, R_{\text{blank}}$: sample and blank readings; $d\theta/dc_s$: concentration-graph slope.

Kinetic theory & molecular speeds

T: 4-12

Topic symbols. P, V, T: pressure/volume/temperature; n, N: moles/molecule count; m, M: molecule mass/molar mass; R, k_B, N_A: gas, Boltzmann, Avogadro constants; ρ: density; c: molecular speed.

Molecular speed averages

$$\bar{c} = \frac{1}{N}\sum_{i=1}^{N} c_i, \quad \overline{c^2} = \frac{1}{N}\sum_{i=1}^{N} c_i^2, \quad c_{\text{rms}} = \sqrt{\overline{c^2}}$$

$$\overline{c_x^2} = \overline{c_y^2} = \overline{c_z^2} = \frac{1}{3}\overline{c^2}$$

c_i: speed of molecule i; $\bar{c}$: mean speed; c_{rms}: root-mean-square speed; c_x, c_y, c_z: velocity components. Overbar denotes ensemble mean; component equality assumes isotropic equilibrium.

Wall collision: impulse, interval & mean force

$$\Delta p_x = 2m|c_x|, \quad \Delta t = \frac{2\ell}{|c_x|}, \quad F_i = \frac{mc_x^2}{\ell}$$

Δp_x: momentum transferred per elastic collision; Δt: interval between hits on the same wall; ℓ: cube side; F_i: mean force from one molecule. m: molecular mass; other symbols defined in this topic's formula groups.

Relative molecular speed & collision-time identities

$$\bar{c}_{\text{rel}} = \sqrt{2}\,\bar{c}, \quad z_c = \frac{\bar{c}}{\lambda_{\text{mfp}}}, \quad \tau_c = \frac{\lambda_{\text{mfp}}}{\bar{c}}$$

d: hard-sphere molecular diameter; σ_c: collision cross section; λ_{mfp}: mean free path; n_N: number density; $\bar{c}_{\text{rel}}$: mean relative speed of identical molecules; z_c: collision rate per molecule; τ_c: mean collision time. Stationary-target approximation omits $\sqrt{2}$.
Meaning: mean free path is the mean distance travelled between collisions, not the distance between neighbouring molecules.

Maxwell distributions

T: 7-12

Topic symbols. c, c_x, c_y, c_z: speed/velocity components

One velocity component & three-dimensional density

$$f_x(c_x) = \sqrt{\frac{m}{2\pi k_B T}}\exp\left(-\frac{mc_x^2}{2k_B T}\right)$$

$$dN = N\left(\frac{m}{2\pi k_B T}\right)^{3/2} e^{-m(c_x^2+c_y^2+c_z^2)/(2k_B T)}dc_x\,dc_y\,dc_z$$

f_x: normalized component probability density; c_x, c_y, c_z: velocity components; m: molecule mass; k_B: Boltzmann constant; T: absolute temperature; N: total count; dN: count in the displayed velocity-space element.

Speed distribution & moments

$$f(c) = 4\pi\left(\frac{m}{2\pi k_B T}\right)^{3/2} c^2 e^{-mc^2/(2k_B T)}$$

$$dN = Nf(c)\,dc, \quad \int_0^\infty f(c)\,dc = 1, \quad \langle c^q\rangle = \int_0^\infty c^q f(c)\,dc$$

$c \geq 0$: molecular speed; $f(c)$: normalized speed density (s m^{-1}); dc: speed interval; q: moment exponent; $\langle c^q\rangle$: moment. A number-distribution graph $Nf(c)$ has total area N, whereas $f(c)$ has area 1.

Mean speed & speed ratios

$$\bar{c} = \sqrt{\frac{8k_B T}{\pi m}}, \qquad c_{\text{mp}} : \bar{c} : c_{\text{rms}} = 1 : \frac{2}{\sqrt{\pi}} : \sqrt{\frac{3}{2}}$$

$$\bar{c} = \frac{2}{\sqrt{\pi}}c_{\text{mp}}, \quad c_{\text{rms}} = \sqrt{\frac{3}{2}}c_{\text{mp}}$$

c_{mp}: speed at the peak of $f(c)$; $\bar{c}$: mean speed; c_{rms}: rms speed. Replace k_B/m by R/M for molar forms; R: gas constant; M: molar mass.

Meaning: most probable speed is the distribution peak; mean speed is the ordinary average; rms speed is the energy-related average.

Translational kinetic-energy distribution

$$\varepsilon = \frac{1}{2}mc^2, \quad g(\varepsilon) = \frac{2}{\sqrt{\pi}(k_B T)^{3/2}}\sqrt{\varepsilon}\,e^{-\varepsilon/(k_B T)}$$

$$dN = Ng(\varepsilon)d\varepsilon, \quad \varepsilon_{\text{mp}} = \frac{1}{2}k_B T, \quad \bar{\varepsilon} = \frac{3}{2}k_B T$$

ε: kinetic energy per molecule (J); g: energy probability density (J^{-1}); ε_{mp}: peak of the energy density. Energy at the most probable speed is $mc_{\text{mp}}^2/2 = k_B T$; it differs from the most probable energy.

Gaussian integrals used in the derivations

$$\int_0^\infty x^q e^{-ax^2}dx = \frac{\Gamma((q+1)/2)}{2a^{(q+1)/2}} \quad (a>0,\ q>-1)$$

$$\int_0^\infty e^{-ax^2}dx = \frac{\sqrt{\pi}}{2\sqrt{a}}, \quad \int_0^\infty x^2 e^{-ax^2}dx = \frac{\sqrt{\pi}}{4a^{3/2}}$$

$$\int_0^\infty x^3 e^{-ax^2}dx = \frac{1}{2a^2}, \quad \int_0^\infty x^4 e^{-ax^2}dx = \frac{3\sqrt{\pi}}{8a^{5/2}}$$

x: integration variable; a: positive Gaussian coefficient; q: real power; Γ: gamma function, $\Gamma(z+1) = z\Gamma(z)$, $\Gamma(1/2) = \sqrt{\pi}$. These are mathematical tools, not additional gas laws.

Equipartition & heat capacities

T: 13-16

Topic symbols. n: moles; T: temperature; R, k_B: gas/Boltzmann constants; $C_{V,m}, C_{P,m}$: molar capacities; $\gamma = C_{P,m}/C_{V,m}$.

Coordinates, constraints & vibration counts

$$f_{\text{coord}} = 3N_a - q, \quad N_{\text{vib}} = 3N_a - 5 \text{ (linear)}, \quad N_{\text{vib}} = 3N_a - 6 \text{ (nonlinear)}$$

$$f_E = f_t + f_r + 2N_{\text{vib,active}}$$

N_a: atoms per molecule (or particles in the constrained system); q: independent coordinate constraints; f_{coord}: independent coordinates; N_{vib}: normal vibration-mode count; f_E: active quadratic energy-term count; $f_t = 3$: translations; $f_r = 2$ linear / 3 nonlinear rotations. Each active harmonic vibration contributes two energy terms.

Real gases & critical constants

T: 16-20; 48

Topic symbols. $v_m = V/n$: molar volume

Equation of state & excluded volume

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT, \quad P = \frac{RT}{v_m - b} - \frac{a}{v_m^2}$$

$$v_m = \frac{V}{n}, \quad b = 4N_A\frac{4\pi r^3}{3} = \frac{2\pi}{3}N_A d^3$$

P, V, T: pressure, total volume, temperature; n: moles; v_m: molar volume; a: attraction parameter (Pa m^6 mol^{-2}); b: excluded molar volume (m^3 mol^{-1}); r: molecular radius; $d = 2r$: diameter; N_A: Avogadro constant; R: gas constant.

Critical-point conditions & derivatives

$$\left(\frac{\partial P}{\partial v_m}\right)_{T_c} = 0, \quad \left(\frac{\partial^2 P}{\partial v_m^2}\right)_{T_c} = 0 \quad (v_m = v_c)$$

$$\left(\frac{\partial P}{\partial v_m}\right)_T = -\frac{RT}{(v_m - b)^2} + \frac{2a}{v_m^3}$$

$$\left(\frac{\partial^2 P}{\partial v_m^2}\right)_T = \frac{2RT}{(v_m - b)^3} - \frac{6a}{v_m^4}$$

T_c: critical temperature; v_c: critical molar volume; other symbols defined in this topic's formula groups. Conditions identify the horizontal inflection on the model critical isotherm.

Van der Waals critical constants & parameter recovery

$$v_c = 3b, \quad P_c = \frac{a}{27b^2}, \quad T_c = \frac{8a}{27Rb}$$

$$b = \frac{RT_c}{8P_c}, \quad a = \frac{27R^2T_c^2}{64P_c}, \quad Z_c = \frac{3}{8}$$

P_c, T_c, v_c: critical model values; Z_c: critical compressibility factor; M: molar mass; ρ_c: critical density. These van der Waals predictions need not match experimental critical values exactly.

Meaning: the critical constants locate the model point where liquid and vapour cease to be distinct phases.

Reduced equation & Boyle temperature

$$P_r = \frac{P}{P_c}, \quad v_r = \frac{v_m}{v_c}, \quad T_r = \frac{T}{T_c}$$

$$\left(P_r + \frac{3}{v_r^2}\right)(3v_r - 1) = 8T_r, \quad T_B = \frac{a}{Rb} = \frac{27}{8}T_c$$

P_r, v_r, T_r: dimensionless reduced variables; T_B: van der Waals Boyle temperature (zero second virial coefficient). a, b, R, T_c defined in this topic's formula groups.

Internal energy of the model

$$U_m(T, v_m) = C_{V,m}T - \frac{a}{v_m} + U_{m,0}, \qquad \left(\frac{\partial U_m}{\partial v_m}\right)_T = \frac{a}{v_m^2}$$

U_m: molar internal energy; $U_{m,0}$: reference constant; $C_{V,m}$: constant molar heat capacity. The integrated form assumes $C_{V,m}$ constant; a/v_m^2 is the model internal pressure.

First law & thermodynamic processes

T: 20-28

Topic symbols. P, V, T: pressure/volume/temperature; n: moles; R: gas constant; Q: heat in; W: work by gas; U, H: energy/enthalpy; $C_{V,m}, C_{P,m}$: molar capacities.

Free expansion of an isolated gas

$$Q = W = 0 \implies \Delta U = 0$$

$Q_{\text{net}}, W_{\text{net}}$: signed cycle totals; P_H, P_L: high/low rectangle pressures; V_H, V_L: large/small volumes. Rectangle formula positive for clockwise traversal. Ideal-gas free expansion also gives $\Delta T = 0$.

Adiabatic processes & steady flow

T: 23-28

Topic symbols. P, V, T: pressure/volume/temperature; γ: capacity ratio; W: work by gas; w: flow speed.

Adiabatic work & PV slopes

$$Q = 0, \quad W_{\text{by}} = -\Delta U = \frac{P_1V_1 - P_2V_2}{\gamma - 1} = \frac{nR(T_1 - T_2)}{\gamma - 1}$$

$$W_{\text{by}} = nC_{V,m}(T_1 - T_2), \quad \left(\frac{\partial P}{\partial V}\right)_T = -\frac{P}{V}, \quad \left(\frac{\partial P}{\partial V}\right)_S = -\gamma\frac{P}{V}$$

Q: heat in; U: internal energy; n: moles; R: gas constant; $C_{V,m}$: molar constant-volume capacity; S: entropy. Slopes compared at the same (P, V); reversible adiabatic is isentropic.

Enthalpy & steady-flow energy balance

$$h = u + Pv, \qquad h_1 + \frac{w_1^2}{2} + gz_1 + q = h_2 + \frac{w_2^2}{2} + gz_2 + w_s$$

h: specific enthalpy (J kg^{-1}); u: specific internal energy; v: specific volume (m^3 kg^{-1}); w_1, w_2: inlet/outlet flow speeds (m s^{-1}); z_1, z_2: elevations; g: gravity; q: heat input per mass; w_s: shaft-work output per mass. Steady, one-inlet/one-outlet flow.

Heat engines & Carnot cycle

T: 29-37

Topic symbols. $Q_H, Q_L > 0$: hot heat input/cold heat rejection; W: cycle work; T_H, T_L: hot/cold temperatures; γ: capacity ratio.

Four Carnot legs: work BY the ideal gas

$$W_{AB} = nRT_H \ln\frac{V_B}{V_A}, \quad W_{BC} = nC_{V,m}(T_H - T_L)$$

$$W_{CD} = nRT_L \ln\frac{V_D}{V_C}, \quad W_{DA} = nC_{V,m}(T_L - T_H)$$

$$\frac{V_B}{V_A} = \frac{V_C}{V_D}, \quad W_{\text{cycle}} = nR(T_H - T_L)\ln\frac{V_B}{V_A}$$

A, B, C, D: sequential cycle corners; AB: hot isothermal expansion; BC: adiabatic expansion; CD: cold isothermal compression; DA: adiabatic compression. n: moles; $C_{V,m}$: molar capacity; R: gas constant; V_X: corner volume. Compression work is negative.

Carnot efficiency sensitivity

$$\frac{\partial \eta_C}{\partial T_H} = \frac{T_L}{T_H^2}, \quad \frac{\partial \eta_C}{\partial T_L} = -\frac{1}{T_H}$$

$$\Delta\eta_{H\uparrow} \simeq \frac{T_L}{T_H^2}\delta T, \quad \Delta\eta_{L\downarrow} = \frac{\delta T}{T_H}$$

$\delta T > 0$: equal small source increase or sink decrease; prime: new condition; η_C: Carnot efficiency. Source-change expression with $\simeq$ is a small-change approximation; fixed-T_H sink-change expression is exact.

Refrigerators & entropy laws

T: 34-40

Topic symbols. $Q_H, Q_L > 0$: heat magnitudes; COP: heat moved per work input.

Heat-pump coefficient of performance

$$\mathrm{COP}_{HP} = \frac{Q_H}{W_{\mathrm{in}}} = \mathrm{COP}_R + 1, \quad \mathrm{COP}_{HP,C} = \frac{T_H}{T_H - T_L}$$

$Q_L > 0$: heat extracted from cold body; $Q_H > 0$: heat delivered to hot body; W_{in}: input work; COP: coefficient of performance; subscripts R, HP, C: refrigerator, heat pump, reversible Carnot limit. T_H, T_L: temperatures in K.
Meaning: refrigerator COP is heat removed from the cold space per unit input work; it can exceed one.

Entropy definition & cyclic identities

$$dS = \frac{\delta Q_{\mathrm{rev}}}{T}, \quad S_2 - S_1 = \int_1^2 \frac{\delta Q_{\mathrm{rev}}}{T}$$

$$\oint dS = 0, \quad \oint \frac{\delta Q_{\mathrm{rev}}}{T} = 0, \quad \oint \frac{\delta Q}{T_b} \le 0$$

S: entropy (J K^{-1}); δQ_{rev}: reversible heat increment into system; T: reversible-path temperature; T_b: boundary/reservoir temperature for actual heat transfer. Entropy returns after any system cycle; the heat integral inequality distinguishes irreversible cycles.

Entropy balance & generation

$$\Delta S_{\mathrm{sys}} = \int \frac{\delta Q}{T_b} + S_{\mathrm{gen}}, \quad S_{\mathrm{gen}} \ge 0$$

$$\Delta S_{\mathrm{universe}} = \Delta S_{\mathrm{sys}} + \Delta S_{\mathrm{surr}} \ge 0$$

S_{gen}: entropy generated by irreversibility; sys: system; surr: surroundings. Closed system, heat exchange at defined boundary temperatures. Isolated: $\Delta S = S_{\mathrm{gen}}$; reversible: $S_{\mathrm{gen}} = 0$. Reversible alone does not imply $\Delta S_{\mathrm{sys}} = 0$.

Meaning: entropy generation measures irreversibility; it vanishes for a reversible process.

Reservoir heat transfer & unavailable work

$$\Delta S_{\mathrm{hot}} = -\frac{Q}{T_H}, \quad \Delta S_{\mathrm{cold}} = \frac{Q}{T_L}$$

$$\Delta S_{\mathrm{total}} = Q\left(\frac{1}{T_L} - \frac{1}{T_H}\right) > 0, \quad W_{\mathrm{lost}} = T_0 S_{\mathrm{gen}}$$

$Q > 0$: heat transferred hot to cold; $T_H > T_L$: fixed reservoir temperatures; T_0: environment temperature; W_{lost}: destroyed available work. The work-loss formula uses entropy generation, not an arbitrary system entropy increase.

Entropy changes & phase transitions

T: 40-43

Topic symbols. m_g: mass; L: specific latent heat.

Ideal gas: entropy in temperature & volume

$$\Delta S = nC_{V,m} \ln\frac{T_2}{T_1} + nR \ln\frac{V_2}{V_1}$$

ΔS: total entropy change; n: moles; $C_{V,m}, C_{P,m}$: constant molar capacities; R: gas constant; P_i, V_i, T_i: endpoint states. Applies to any path between equilibrium ideal-gas states; use a reversible path for evaluating entropy.

Constant volume, constant pressure & isothermal

$$\Delta S_V = n\int_{T_1}^{T_2} \frac{C_{V,m}(T)}{T}\,dT, \quad \Delta S_P = n\int_{T_1}^{T_2} \frac{C_{P,m}(T)}{T}\,dT$$

$$T = \mathrm{const}: \quad \Delta S = nR \ln\frac{V_2}{V_1} = -nR \ln\frac{P_2}{P_1}$$

Subscripts V, P: fixed volume/pressure; $C(T)$: temperature-dependent molar capacity. For a general body with mass m_g and constant specific heat c_s, $\Delta S = m_g c_s \ln(T_2/T_1)$.

Variable heat capacity: temperature-volume & PV forms

$$C_{P,m} = a + \kappa T, \quad C_{V,m} = b + \kappa T, \quad a - b = R$$

$$\Delta S = n\left[b \ln\frac{T_2}{T_1} + \kappa(T_2 - T_1) + R \ln\frac{V_2}{V_1}\right]$$

$$= n\left[b \ln\frac{P_2}{P_1} + a \ln\frac{V_2}{V_1} + \kappa(T_2 - T_1)\right]$$

a, b: molar-capacity intercepts (J mol^{-1} K^{-1}); κ: common slope (J mol^{-1} K^{-2}). For mass-specific versions replace $C_{P,m}, C_{V,m}, R, n, \Delta S$ by $c_p, c_v, R_s, 1, \Delta s$ consistently.

Thermodynamic potentials

T: 43-45; 49

Topic symbols. composition held fixed.

Internal energy

$$U = U(S,V), \quad dU = T\,dS - P\,dV$$
$$T = \left(\frac{\partial U}{\partial S}\right)_V, \qquad P = -\left(\frac{\partial U}{\partial V}\right)_S$$

U: internal energy (J); S: entropy (J K^{-1}); V: volume (m^3); T: absolute temperature; P: pressure. All potential identities on this page assume fixed composition, a simple compressible equilibrium system and only PV work.

Enthalpy

$$H = U + PV = H(S,P), \quad dH = T\,dS + V\,dP$$
$$T = \left(\frac{\partial H}{\partial S}\right)_P, \qquad V = \left(\frac{\partial H}{\partial P}\right)_S$$

H: enthalpy (J); other symbols defined in this topic's formula groups. $\delta Q_P = dH$ for a constant-pressure process with only PV work and appropriate mechanical balance.

Helmholtz free energy

$$F = U - TS = F(T,V), \quad dF = -S\,dT - P\,dV$$
$$S = -\left(\frac{\partial F}{\partial T}\right)_V, \qquad P = -\left(\frac{\partial F}{\partial V}\right)_T$$

F: Helmholtz free energy (J); U, T, S, P, V defined in this topic's formula groups. $W_{\text{by,rev}} = -\Delta F$ on an isothermal reversible path with only PV work.

Meaning: Helmholtz energy is the natural energy potential for temperature and volume.

Gibbs free energy

$$G = H - TS = U + PV - TS = F + PV = G(T,P)$$
$$dG = -S\,dT + V\,dP, \quad S = -\left(\frac{\partial G}{\partial T}\right)_P, \quad V = \left(\frac{\partial G}{\partial P}\right)_T$$

G: Gibbs free energy (J); H, F, U, T, S, P, V defined in this topic's formula groups. For coexisting phases at fixed T, P, equilibrium requires equal molar Gibbs free energy, not equal total G for arbitrary phase amounts.

Meaning: Gibbs energy is the natural potential for temperature and pressure; equal molar/specific values identify phase equilibrium.

Gibbs-Helmholtz identities

$$H = G - T\left(\frac{\partial G}{\partial T}\right)_P, \quad U = F - T\left(\frac{\partial F}{\partial T}\right)_V$$
$$\left(\frac{\partial (G/T)}{\partial T}\right)_P = -\frac{H}{T^2}, \qquad \left(\frac{\partial (F/T)}{\partial T}\right)_V = -\frac{U}{T^2}$$

H, G, U, F: enthalpy, Gibbs, internal and Helmholtz energies; T: absolute temperature. Partial-derivative subscripts identify the variable held fixed.

Maxwell's thermodynamic relations

T: 45-47

Topic symbols. subscripts indicate held-fixed variables.

From internal energy U(S,V)

$$\boxed{\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V}$$

T: temperature; V: volume; S: entropy; P: pressure; U: internal energy. Fixed amount/-composition; simple compressible equilibrium system.

From Helmholtz energy F(T,V)

$$\boxed{\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V}$$

F: Helmholtz energy; T, V, S, P defined in this topic's formula groups. Mixed derivatives of a smooth state potential commute.

From enthalpy H(S,P)

$$\boxed{\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P}$$

H: enthalpy; other variables defined in this topic's formula groups. A subscript specifies what stays constant in the derivative.

From Gibbs energy G(T,P)

$$\boxed{\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P}$$

G: Gibbs energy; T, V, S, P defined in this topic's formula groups.

Parent equation & examined special identities

$$\frac{\partial T}{\partial x}\frac{\partial S}{\partial y} - \frac{\partial T}{\partial y}\frac{\partial S}{\partial x} = \frac{\partial P}{\partial x}\frac{\partial V}{\partial y} - \frac{\partial P}{\partial y}\frac{\partial V}{\partial x}$$

$$\left(\frac{\partial P}{\partial T}\right)_S \left(\frac{\partial V}{\partial S}\right)_T - \left(\frac{\partial P}{\partial S}\right)_T \left(\frac{\partial V}{\partial T}\right)_S = 1$$

$$\left(\frac{\partial T}{\partial P}\right)_V \left(\frac{\partial S}{\partial V}\right)_P - \left(\frac{\partial T}{\partial V}\right)_P \left(\frac{\partial S}{\partial P}\right)_V = 1$$

x, y: any independent state-coordinate pair; each x derivative holds y fixed and each y derivative holds x fixed. T, S, P, V defined in this topic's formula groups. The last two identities are choices of coordinates in the parent relation.

Use: the parent identity generates Maxwell relations by choosing the independent state variables.

TdS equations & capacity difference

T: 22-23; 40; 47-49

Topic symbols. subscript m: molar quantity

Heat capacities & entropy derivatives

$$C_V = T\left(\frac{\partial S}{\partial T}\right)_V, \quad C_P = T\left(\frac{\partial S}{\partial T}\right)_P$$

$$TdS = C_V\,dT + T\left(\frac{\partial P}{\partial T}\right)_V dV$$

$$TdS = C_P\,dT - T\left(\frac{\partial V}{\partial T}\right)_P dP$$

C_V, C_P: total heat capacities (J K^{-1}); T: temperature; S: total entropy; P, V: pressure and total volume. For molar versions use $C_{V,m}, C_{P,m}, S_m, v_m$ consistently.

General energy & capacity identities

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial P}{\partial T}\right)_V - P$$

$$C_P - C_V = \left[\left(\frac{\partial U}{\partial V}\right)_T + P\right]\left(\frac{\partial V}{\partial T}\right)_P$$

$$= T\left(\frac{\partial P}{\partial T}\right)_V \left(\frac{\partial V}{\partial T}\right)_P = -T\left(\frac{\partial V}{\partial T}\right)_P^2 \left(\frac{\partial P}{\partial V}\right)_T$$

U: internal energy; T, P, V, C_P, C_V defined in this topic's formula groups. Stable single phase: $(\partial P/\partial V)_T < 0$, giving $C_P \geq C_V$.

Partial-derivative reciprocity & cyclic rule

$$\left(\frac{\partial x}{\partial y}\right)_z = \frac{1}{(\partial y/\partial x)_z}, \qquad \left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1$$

x, y, z: related state variables; derivatives assumed finite and invertible where used.

Van der Waals heat-capacity difference

$$\left(\frac{\partial P}{\partial T}\right)_{v_m} = \frac{R}{v_m - b}, \quad \left(\frac{\partial v_m}{\partial T}\right)_P = \frac{R/(v_m - b)}{RT/(v_m - b)^2 - 2a/v_m^3}$$

$$C_{P,m} - C_{V,m} = \frac{R}{1 - 2a(v_m - b)^2/(RTv_m^3)}$$

$$\simeq \frac{R}{1 - 2a/(RTv_m)} \simeq R\left(1 + \frac{2a}{RTv_m}\right)$$

v_m: molar volume; a, b: van der Waals parameters; R: gas constant; T: temperature; $C_{P,m}, C_{V,m}$: molar capacities. First approximation: $b \ll v_m$; second also $2a/(RTv_m) \ll 1$. The final expression is not exact.

Use: computes how real-gas attraction changes the difference between constant-pressure and constant-volume heat capacities.

Van der Waals entropy

$$dS_m = C_{V,m}(T)\frac{dT}{T} + R\frac{dv_m}{v_m - b}$$

$$\Delta S_m = C_{V,m} \ln\frac{T_2}{T_1} + R \ln\frac{v_{m,2} - b}{v_{m,1} - b}, \quad \Delta S = n\Delta S_m$$

S_m: molar entropy; n: moles; $v_{m,i}$: endpoint molar volumes; b: excluded molar volume. Integrated form assumes constant $C_{V,m}$; total-entropy form uses fixed n.

Phase equilibrium & Clapeyron equation

T: 49-54

Topic symbols. T, P: coexistence temperature/pressure; M: molar mass.

Phase equilibrium & transition entropy

$$g_1 = g_2, \quad dg_1 = dg_2, \quad \Delta s = \frac{L}{T}$$

g_1, g_2: specific Gibbs free energies (J kg^{-1}); s_1, s_2: specific entropies (J kg^{-1} K^{-1}); v_1, v_2: specific volumes (m^3 kg^{-1}); L: latent heat from phase 1 to 2 (J kg^{-1}); P_{sat}: coexistence pressure; T: transition temperature.

Meaning: the Clapeyron slope tells how the equilibrium melting or boiling temperature changes with pressure.

Clausius approximation for vaporization

$$v_{m,v} \gg v_{m,l}, \quad v_{m,v} \simeq \frac{RT}{P}$$

$$\frac{dP}{dT} \simeq \frac{L_m P}{RT^2}, \quad \frac{d\ln P}{dT} \simeq \frac{L_m}{RT^2}$$

$v_{m,v}, v_{m,l}$: molar vapour/liquid volumes; L_m: molar latent heat (J mol^{-1}); R: universal gas constant. Vapour ideal; liquid volume neglected. In mass-specific form replace L_m, R by L, R_s.

Part II navigator

Click a topic or use the PDF bookmarks. Source references use the printed pages of the question banks.

01	Simple harmonic motion	20
02	Springs & oscillator energy	20
03	Pendulums & two-body motion	21
04	Superposition & Lissajous figures	21
05	Damped oscillations	22
06	Forced oscillations & resonance	23
07	Electrical oscillations & radiation	23
08	Travelling waves	24
09	Standing waves & wave groups	24
10	Matter waves	24
11	Lenses & chromatic aberration	25
12	Achromatic lens combinations	25
13	Interference & double-source fringes	26
14	Thin films, wedges & biprism	26
15	Newton’s rings	27
16	Single-slit & circular diffraction	27
17	Double slit & diffraction grating	28
18	Resolution & polarization	28
19	Wave plates & optical activity	29
20	Kinetic theory & molecular speeds	29
21	Maxwell distributions	30
22	Equipartition & heat capacities	30
23	Real gases & critical constants	30
24	First law & thermodynamic processes	31
25	Adiabatic processes & steady flow	32
26	Heat engines & Carnot cycle	32
27	Refrigerators & entropy laws	33
28	Entropy changes & phase transitions	33
32	Phase equilibrium & Clapeyron equation	34
33	Constants, units & notation	34

Source key

O: PHY121_Optics_Waves_Solutions(1).pdf
T: PHY121_Thermal_Physics_Solutions(1).pdf

Part I contains derivation results and supporting identities that are not used in the banks’ numerical solutions. Part II contains numerical-use formulae, including results shared with derivations. Classification follows the banks’ Part A / Part B sections; a calculation embedded in a theory question stays with Part I unless its formula also appears in Part B. Corrected or generalized variants follow the formula they support. Constants and conversions are collected at the end of Part II.

Simple harmonic motion

O: 3-10

Topic symbols. v, a: velocity/acceleration

Restoring force from potential energy

$$F = -kx = -\frac{dU}{dx}$$

F: force (N); k: spring stiffness (N m^{-1}); x: displacement from equilibrium (m); U: potential energy (J); m: mass (kg); ω_0: natural angular frequency (rad s^{-1}). Dots denote time derivatives.
Use: connects restoring force to the oscillator equation; stiffness sets the restoring strength.

Displacement, velocity & acceleration

$$x = A\cos(\omega_0 t + \phi), \quad v = -A\omega_0 \sin(\omega_0 t + \phi), \quad a = -\omega_0^2 x$$

A: amplitude (m); t: time (s); ϕ: initial phase (rad); $v = \dot{x}$: velocity (m s^{-1}); $a = \ddot{x}$: acceleration (m s^{-2}).
Use: gives position, velocity and acceleration at any instant during ideal SHM.

Speed, extrema & period

$$v = \pm\omega_0\sqrt{A^2 - x^2}, \quad v_{\max} = \omega_0 A, \quad |a|_{\max} = \omega_0^2 A$$

$$T = \frac{2\pi}{\omega_0} = 2\pi\sqrt{\frac{m}{k}}, \qquad f = \frac{1}{T} = \frac{\omega_0}{2\pi}$$

T: period (s); f: frequency (Hz); $v_{\max}$: maximum speed; $|a|_{\max}$: maximum acceleration magnitude. Other symbols defined in this topic's formula groups.
Use: converts displacement into speed and finds the repeat time and oscillation frequency.

Initial conditions & phase quadrant

$$A = \sqrt{x_0^2 + \left(\frac{v_0}{\omega_0}\right)^2}, \quad \phi = \operatorname{atan2}\left(-\frac{v_0}{\omega_0}, x_0\right)$$

$x_0 = x(0)$: initial displacement; $v_0 = v(0)$: initial velocity. This phase formula uses the cosine convention above; atan2 chooses the correct quadrant.

Potential supplied per unit mass

$$\Psi(x) = \alpha x^2 + \Psi_0 : \qquad a = -\frac{d\Psi}{dx} = -2\alpha x, \quad \omega_0 = \sqrt{2\alpha}$$

Ψ: potential energy per unit mass (J kg^{-1}); α: quadratic coefficient (s^{-2}); Ψ_0: constant offset. For total potential $U = \alpha x^2 + U_0$, instead $\omega_0 = \sqrt{2\alpha/m}$.

Springs & oscillator energy

O: 5-10

Topic symbols. $\omega_0 = 2\pi f$

Kinetic, potential & total energy

$$U = \frac{1}{2}kx^2, \quad K = \frac{1}{2}mv^2 = \frac{1}{2}k(A^2 - x^2)$$

$$E = K + U = \frac{1}{2}kA^2 = \frac{1}{2}m\omega_0^2 A^2 = 2\pi^2 m f^2 A^2$$

U: potential energy; K: kinetic energy; E: total mechanical energy (all J); k: stiffness; m: mass; x: displacement; v: velocity; A: amplitude; f: frequency; $\omega_0 = 2\pi f$.
Use: tracks how energy alternates between motion and spring deformation.

Amplitude from mechanical energy

$$A = \sqrt{\frac{2E}{k}}$$

$\langle\cdot\rangle$: average over one complete cycle. All other symbols defined in this topic's formula groups; ideal undamped SHM.

Uniform spring: cutting & adding load

$$k\ell = \text{constant}, \quad k_{\text{piece}} = k\frac{\ell}{\ell_{\text{piece}}}, \quad \frac{f_2}{f_1} = \sqrt{\frac{m_1}{m_2}} \quad (k \text{ fixed})$$

ℓ: original spring length; ℓ_{piece}: cut-piece length; k_{piece}: its stiffness. m_1, m_2: total moving masses; f_1, f_2: corresponding frequencies. Uniform spring; negligible spring mass.

Spring stiffness from static extension

$$kh = mg, \qquad k = \frac{mg}{h}$$

m_s: uniform spring mass; m: end load; m_{eff}: effective moving mass; T: period; h: static load extension; g: gravitational acceleration. The $m_s/3$ model assumes displacement varies linearly along the spring; it is an approximation.
Use: distinguishes the spring's inertial contribution from its static load extension.

Pendulums & two-body motion

O: 4-9

Topic symbols. ω_0: angular frequency

Physical-pendulum period

$$T = 2\pi\sqrt{\frac{I}{Mgl}}$$

τ: restoring torque (N m); M: body mass; g: gravity; l: pivot-to-centre-of-mass distance; θ: small angular displacement; I: inertia about pivot (kg m^2); I_G: inertia about centre of mass; K_G: radius of gyration; T: period.

Equivalent pendulum length

$$L_{\text{eq}} = \frac{I}{Ml}$$

L_{eq}: equivalent simple-pendulum length; l': centre-of-mass distance to conjugate pivot. Other symbols defined in this topic's formula groups; small oscillations.

Uniform rod suspended at one end

$$I = \frac{1}{3}ML^2, \quad l = \frac{L}{2}, \quad T = 2\pi\sqrt{\frac{2L}{3g}}, \quad L_{\text{eq}} = \frac{2L}{3}$$

L: rod length; M: rod mass; I: inertia about end; l: end-to-centre distance; g: gravity; T: small-oscillation period.

Reduced mass & relative frequency

$$\mu = \frac{m_1 m_2}{m_1 + m_2}, \quad \omega_0 = \sqrt{\frac{k}{\mu}}, \quad f = \frac{1}{2\pi}\sqrt{\frac{k}{\mu}}, \quad T = 2\pi\sqrt{\frac{\mu}{k}}$$

x_1, x_2: mass coordinates; ℓ_0: natural spring length; x: relative extension; m_1, m_2: masses; μ: reduced mass; k: stiffness; T: relative-motion period. Frictionless horizontal motion.
Meaning: reduced mass converts relative motion of two bodies into an equivalent one-body oscillator.

Superposition & Lissajous figures

O: 11-14; 38

Perpendicular SHMs: equal frequencies

$$x = a\sin(\omega t + \delta), \quad y = b\sin\omega t$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab}\cos\delta = \sin^2\delta$$

x, y: perpendicular coordinates; a, b: their amplitudes; ω: angular frequency; δ: phase difference; t: time.
Use: predicts the path traced by two perpendicular oscillations, including tilted ellipses.

Line, ellipse & circle conditions

$$\delta = 0: \; y = \frac{b}{a}x; \qquad \delta = \pi: \; y = -\frac{b}{a}x$$

$$\delta = \frac{\pi}{2} \text{ or } \frac{3\pi}{2}: \; \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1; \qquad a = b: \; x^2 + y^2 = a^2$$

Equal component frequencies are required. x, y, a, b, δ defined in this topic's formula groups. With quadrature phase and equal amplitudes the ellipse becomes a circle.

Figure-of-eight trajectory

$$\phi = 0: \qquad \frac{y^2}{b^2} = \frac{4x^2}{a^2}\left(1 - \frac{x^2}{a^2}\right)$$

ϕ: phase offset in the doubled-frequency motion; a, b: amplitudes; x, y: coordinates; ω: lower angular frequency. Physical trajectory: $|x| \le a$, $|y| \le b$.

Frequency measurement by tangencies

$$\frac{f_x}{f_y} = \frac{n_v}{n_h}$$

f_x, f_y: horizontal and vertical frequencies; n_v: tangencies on one vertical bounding side; n_h: tangencies on one horizontal side. Use a stationary, nondegenerate closed figure; retraced curves can make counting ambiguous.

Damped oscillations

O: 15-19

Topic symbols. $\omega_0 = \sqrt{k/m}$

Natural frequency & damping-rate parameters

$$\omega_0^2 = \frac{k}{m}, \quad \gamma_d = \frac{b}{2m}, \quad p = \frac{b}{m} = 2\gamma_d$$

m: mass; b: viscous damping constant (kg s^{-1}); k: stiffness; x: displacement; γ_d: amplitude decay rate (s^{-1}); p: source damping parameter; $r_\pm$: characteristic exponents.

Meaning: damping rate measures how rapidly motion decays; the characteristic roots determine whether oscillation is possible.

Damped frequency, period & amplitude envelope

$$\omega_d = \sqrt{\omega_0^2 - \gamma_d^2}, \quad T_d = \frac{2\pi}{\omega_d}, \quad A(t) = A_0 e^{-\gamma_d t}$$

ω_d: damped angular frequency; T_d: damped period; A_0: initial amplitude envelope; $A(t)$: envelope; ϕ: phase constant; t: time. $x(0) = A_0 \cos\phi$, so initial displacement need not equal A_0.

Meaning: underdamping produces repeated crossings of equilibrium while the amplitude shrinks exponentially.

Amplitude mean lifetime & reduction time

$$\tau_A = \frac{1}{\gamma_d} = \frac{2m}{b}, \quad t = \frac{1}{\gamma_d} \ln \frac{A_0}{A(t)}, \quad t_{1/2,A} = \frac{\ln 2}{\gamma_d}$$

τ_A: amplitude e-folding time; N: number of periods; A_j, A_{j+N}: successive same-sign peak magnitudes separated by N periods; Λ: log decrement per period; $t_{1/2,A}$: amplitude half-life.

Meaning: amplitude mean lifetime is the time for the amplitude envelope to become 1/e (about 36.8 percent) of its starting value. A half-life refers to 50 percent, so it is different. Logarithmic decrement measures the amplitude loss per cycle.

Energy lifetime, quality factor & decay

$$\overline{E} \simeq E_0 e^{-2\gamma_d t}, \quad E_0 \simeq \frac{1}{2} k A_0^2$$

$$\tau_E = \frac{1}{2\gamma_d} = \frac{m}{b}, \quad Q = \frac{\omega_0}{2\gamma_d} = \frac{\sqrt{mk}}{b}, \quad t_{1/2,E} \simeq \tau_E \ln 2$$

E_{inst}: exact instantaneous mechanical energy; $\overline{E}$: slowly varying cycle-mean energy; E_0: initial mean energy; τ_E: energy e-folding time; Q: dimensionless quality factor; P_{loss}: dissipated power; ΔE_{cycle}: energy lost per cycle. Energy-envelope and cycle-loss expressions use weak damping.

Meaning: quality factor Q measures how weakly an oscillator loses energy relative to the energy it stores. Larger Q means slower relative loss and sharper resonance. Energy lifetime is the time for the energy envelope to fall to 1/e; it is half the amplitude lifetime.

Forced oscillations & resonance

O: 20-24

Topic symbols. $\omega_0 = \sqrt{k/m}$; $\gamma_d = b/(2m)$

Steady-state displacement amplitude

$$A(\Omega) = \frac{F_0}{\sqrt{(k - m\Omega^2)^2 + b^2\Omega^2}} = \frac{F_0/m}{\sqrt{(\omega_0^2 - \Omega^2)^2 + 4\gamma_d^2\Omega^2}}$$

m: mass; b: damping constant; k: stiffness; F_0: force amplitude (N); Ω: drive angular frequency; t: time; x_{ss}: steady displacement; A: its amplitude; $\omega_0 = \sqrt{k/m}$; $\gamma_d = b/(2m)$.

Use: predicts how strongly an oscillator responds after its initial transient has died away.

Amplitude at the natural frequency

$$A(\omega_0) = \frac{F_0}{b\omega_0}$$

Ω_r: frequency of maximum displacement amplitude; ω_d: free damped angular frequency. For weaker damping, $\Omega_r \simeq \omega_0$ and $A_{\text{max}} \simeq F_0/(b\omega_0)$. If $\gamma_d \geq \omega_0/\sqrt{2}$, no nonzero-frequency displacement peak.

Use: locates the true maximum of displacement amplitude. It need not occur exactly at the natural frequency.

Normalized amplitude & bandwidth

$$r = \frac{\Omega}{\omega_0}, \quad Q = \frac{m\omega_0}{b}, \quad \frac{A(\Omega)}{A(\omega_0)} = \frac{1/Q}{\sqrt{(1 - r^2)^2 + r^2/Q^2}}$$

$$\frac{A(\Omega)}{A_{\text{max}}} = \frac{\sqrt{1 - 1/(4Q^2)}}{Q\sqrt{(1 - r^2)^2 + r^2/Q^2}}, \quad Q > \frac{1}{\sqrt{2}}$$

r: frequency ratio; Q: quality factor; A_{max}: exact peak amplitude. The source numerical ratio uses $A(\omega_0)$ as the approximate resonance amplitude.

Use: compares the response at a chosen drive frequency with a reference resonance amplitude.

Electrical oscillations & radiation

O: 22-24

Topic symbols. L, R, C: inductance, resistance, capacitance; q, i: charge/current; $\mathcal{E}_0$: emf amplitude; Ω: drive frequency; t: time

RLC natural frequency, discharge frequency & Q

$$\omega_0 = \frac{1}{\sqrt{LC}}, \quad f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$\omega_d = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}, \quad f_d = \frac{\omega_d}{2\pi}, \quad Q = \frac{\omega_0 L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

ω_0, f_0: undamped resonance frequencies; ω_d, f_d: free damped discharge frequencies; Q: quality factor. Oscillatory discharge requires $R < 2\sqrt{L/C}$. At current resonance $Z = R$, $\varphi_i = 0$.

Meaning: RLC quality factor describes energy retention and resonance selectivity in the circuit.

Radiation law used in the source numerical

$$P_{\text{rad}} = \frac{K_r e^2 a^2}{c^3}, \quad x = A\sin\omega t, \quad a = -\omega^2 A\sin\omega t$$

$$\langle P_{\text{rad}} \rangle = \frac{K_r e^2 \omega^4 A^2}{2c^3}, \quad E_{\text{cycle}} = \frac{\langle P_{\text{rad}} \rangle}{f} = \frac{8\pi^4 K_r e^2 f^3 A^2}{c^3}$$

P_{rad}: radiated power; K_r: coefficient supplied in the question (6×10^9 N m^2 C^{-2}); e: electron charge magnitude; a: acceleration; c: vacuum light speed; A: amplitude; $\omega = 2\pi f$: angular frequency; f: frequency; E_{cycle}: radiated energy per cycle. Nonrelativistic source model.

Travelling waves

O: 24-30

Topic symbols. $k_w = 2\pi/\lambda$: wavenumber; u: particle velocity.

Wave parameters & equation

$$y = A\sin(\omega t - k_w x + \phi), \quad \omega = 2\pi f, \quad k_w = \frac{2\pi}{\lambda}$$

$$v_p = f\lambda = \frac{\omega}{k_w} = \frac{\lambda}{T}, \quad T = \frac{1}{f}$$

y: transverse or longitudinal displacement (m); A: amplitude; x: position; t: time; ϕ: initial phase; f: frequency; ω: angular frequency; k_w: angular wavenumber (rad m^{-1}); λ: wavelength; v_p: phase speed; T: period. Minus sign: propagation in $+x$; plus sign: in $-x$.

Meaning: phase velocity follows a fixed phase such as a crest; it differs from the velocity of an individual medium particle.

Phase differences & distances

$$|\Delta\phi| = k_w|\Delta x| = \frac{2\pi|\Delta x|}{\lambda}, \quad |\Delta x| = \frac{\lambda|\Delta\phi|}{2\pi}$$

$$|\Delta\phi| = \omega|\Delta t| = \frac{2\pi|\Delta t|}{T} \quad (x \text{ fixed})$$

Δx: spatial separation at one time; Δt: time interval at one point; $\Delta\phi$: phase separation in radians. Other symbols defined in this topic's formula groups.

Mean total energy density, intensity & power

$$\langle e\rangle = \frac{1}{2}\rho\omega^2A^2 = 2\pi^2\rho f^2A^2$$

$$I = \langle e\rangle v_p = \frac{1}{2}\rho v_p\omega^2A^2, \qquad P = IS$$

e_K, e_U, e: kinetic, elastic and total energy per volume (J m^{-3}); ρ: medium density (kg m^{-3}); I: mean intensity (W m^{-2}); P: mean power; S: cross-sectional area. Linear harmonic mechanical plane wave; these are cycle averages.

Meaning: energy density is energy per volume; intensity is mean power crossing unit area.

Standing waves & wave groups

O: 26-31

Topic symbols. $k_w = 2\pi/\lambda$: wavenumber; λ: wavelength; v_p, v_g: phase/group velocity

Equal opposite waves: free-end origin

$$y_1 = A\sin(\omega t - k_w x), \quad y_2 = A\sin(\omega t + k_w x)$$

$$y = 2A\cos(k_w x)\sin\omega t, \quad A_x = 2A|\cos k_w x|$$

$$x_{\text{node}} = \frac{(2j+1)\lambda}{4}, \quad x_{\text{antinode}} = \frac{j\lambda}{2}$$

A: each travelling-wave amplitude; y_1, y_2: component displacements; y: resultant; A_x: local amplitude; $k_w = 2\pi/\lambda$; ω: angular frequency; j: integer indexing positions; x, t: position and time.

Fixed-end origin & node spacings

$$y_2 = -A\sin(\omega t + k_w x), \quad y = -2A\sin(k_w x)\cos\omega t$$

$$x_{\text{node}} = \frac{j\lambda}{2}, \quad x_{\text{antinode}} = \frac{(2j+1)\lambda}{4}$$

$$d_{NN} = d_{AA} = \frac{\lambda}{2}, \quad d_{NA} = \frac{\lambda}{4}, \quad L = N_\ell\frac{\lambda}{2}$$

d_{NN}, d_{AA}: consecutive node and antinode spacings; d_{NA}: nearest node-to-antinode distance; L: length spanning N_ℓ complete loops; N_ℓ: loop count. Other symbols defined in this topic's formula groups; boundary placed at $x = 0$.

Matter waves

O: 30-31

Topic symbols. $\hbar = h/(2\pi)$

Nonrelativistic matter wave: source convention

$$\lambda = \frac{h}{mv}, \quad f = \frac{mv^2}{2h}, \quad E = hf = \frac{1}{2}mv^2$$

$$k_w = \frac{mv}{\hbar}, \quad \omega = \frac{\hbar k_w^2}{2m}, \quad v_p = \frac{\omega}{k_w} = \frac{v}{2}, \quad v_g = \frac{d\omega}{dk_w} = v$$

h: Planck constant; $\hbar = h/(2\pi)$; m: particle mass; v: particle speed; E: kinetic energy (rest energy omitted); λ: de Broglie wavelength; f, ω: wave frequencies; k_w: angular wavenumber; v_p, v_g: phase and group velocities.

Use: distinguishes particle/group speed from phase speed in the nonrelativistic kinetic-energy matter-wave convention.

Lenses & chromatic aberration

O: 33-37

Topic symbols. ω_c: dispersive power; D: aperture diameter

Thin lens & lensmaker relation

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right), \quad \mathcal{P} = \frac{1}{f}, \quad M = \frac{v}{u}$$

f: focal length; u, v: signed object and image distances; n: lens refractive index relative to air; R_1, R_2: signed surface radii; $\mathcal{P}$: lens power (m^{-1}); M: signed magnification. Cartesian convention: light travels left to right; radius positive if centre is right of surface.

Two-lens minimum-aberration separation

$$d = f_1 - f_2 \qquad (f_1 > f_2 > 0)$$

R_1, R_2: surface radii of the best-form thin lens for a parallel incident beam; n: refractive index. Second formula: source two-converging-lens arrangement with shared deviation; d: separation; f_1, f_2: focal lengths. This separation is setup-specific.

Plano-convex lens: curved-face radius

$$|R| = (n-1)|f|$$

n_o, n_i: object- and image-space indices; y_o, y_i: corresponding object/image heights; θ_o, θ_i: ray angles to the axis. R: curved-face radius; n: lens index in air; f: focal length. Sine condition accompanies correction of spherical aberration for an aplanatic system.

Achromatic lens combinations

O: 34-37

Topic symbols. R: signed surface radius.

Equivalent focal length

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}, \quad F = \frac{f_1 f_2}{f_1 + f_2 - d}$$

$$d = 0: \quad \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

F: equivalent focal length; f_1, f_2: signed component focal lengths; d: lens separation. Thin lenses in air; distances measured along the optical axis.

Contact achromat: condition & design

$$\frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0, \quad \frac{f_1}{f_2} = -\frac{\omega_1}{\omega_2}$$

$$f_1 = F\frac{\omega_2 - \omega_1}{\omega_2}, \qquad f_2 = -F\frac{\omega_2 - \omega_1}{\omega_1}$$

ω_1, ω_2: component dispersive powers; F: target focal length; f_1, f_2: signed focal lengths. For $F > 0$, $\omega_2 > \omega_1$: lens 1 converges and lens 2 diverges.

Use: pairs converging and diverging powers so the leading colour-dependent focal change cancels.

Separated achromat

$$\frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} - \frac{d(\omega_1 + \omega_2)}{f_1 f_2} = 0$$

$$d = \frac{\omega_1 f_2 + \omega_2 f_1}{\omega_1 + \omega_2}, \qquad \omega_1 = \omega_2: \; d = \frac{f_1 + f_2}{2}$$

d: achromatic separation; f_1, f_2: component focal lengths; ω_1, ω_2: dispersive powers. First-order chromatic correction of equivalent power; fixed separation.

Both source separation conditions

$$d = f_1 - f_2 = \frac{f_1 + f_2}{2} \implies f_1 = 3f_2$$

$$f_1 = 2F, \qquad f_2 = \frac{2F}{3}, \qquad d = \frac{4F}{3}$$

Same material and the specific minimum-spherical-aberration arrangement of the source; F: equivalent focal length; f_1, f_2: positive lens focal lengths; d: separation.

Cemented doublet surface equations

$$\frac{1}{f_1} = (n_1 - 1)\left(\frac{1}{R_1} - \frac{1}{R_c}\right), \quad \frac{1}{f_2} = (n_2 - 1)\left(\frac{1}{R_c} - \frac{1}{R_2}\right)$$

n_1, n_2: crown/flint indices; R_1: first outer radius; R_c: common contact radius; R_2: last outer radius; f_1, f_2: component focal lengths in the thin-lens model. Signed radii use the same left-to-right Cartesian convention.

Interference & double-source fringes

O: 38-41; 44-45

Topic symbols. λ: wavelength; I_s: one-beam intensity; Δ: optical path difference.

Young fringe spacing & wavelength

$$\beta = \frac{\lambda L}{d}, \qquad \lambda = \frac{\beta d}{L}$$

d: source/slit separation; L: source-to-screen distance; x: signed screen coordinate from centre; θ: observation angle; m: integer fringe order; β: fringe width. Small angles, $L \gg d, |x|$; no extra phase offset.

Thin transparent sheet in one path

$$\Delta_{\text{add}} = (n-1)t, \quad x_s = \frac{L}{d}(n-1)t = \frac{\beta}{\lambda}(n-1)t$$

$$N = \frac{x_s}{\beta} = \frac{(n-1)t}{\lambda}, \quad t = \frac{N\lambda}{n-1}, \quad n = 1 + \frac{N\lambda}{t}$$

n: sheet refractive index relative to air; t: thickness; Δ_{add}: added optical path; x_s: magnitude of fringe shift toward sheet; N: number of shifted fringes. d, L, β, λ defined in this topic's formula groups.

Thin films, wedges & biprism

O: 40-45

Reflected-film bright & dark conditions

$$\text{Dark: } 2nt\cos r = m\lambda, \qquad \text{bright: } 2nt\cos r = (m+1/2)\lambda$$

n_0: incident-medium index; n: film index; i, r: incidence/refraction angles; t: film thickness; λ: vacuum wavelength; $m = 0, 1, 2, \ldots$; Δ: propagation optical path difference. These reflection conditions require exactly one phase reversal; transmitted conditions interchange bright and dark.

Minimum thickness & selected wavelengths

$$t_{\text{min,dark}>0} = \frac{\lambda}{2n\cos r}, \quad t_{\text{min,bright}} = \frac{\lambda}{4n\cos r}$$

$$\lambda_{\text{bright}} = \frac{4nt\cos r}{2j-1} \quad (j = 1, 2, \ldots)$$

$t_{\text{min,dark}>0}$: least nonzero dark-film thickness; $t_{\text{min,bright}}$: least bright-film thickness; j: positive integer order. Same one-phase-reversal film model defined in this topic's formula groups.

Wedge fringes

$$t(x) = x\tan\theta \simeq x\theta, \quad 2nt(x) = m\lambda$$

$$\beta = \frac{\lambda}{2n\theta}, \quad \theta \simeq \frac{t_s}{\ell}, \quad t_s = \frac{\lambda\ell}{2n\beta}$$

x: distance from contact edge; θ: small wedge angle (rad); n: gap index; m: dark-fringe order; t_s: spacer thickness at distance ℓ; β: fringe spacing. Normal incidence; one reflection phase reversal; $n = 1$ for air.

Biprism: measured fringe spacing & wavelength

$$L = a + b, \quad \beta = \frac{x_N - x_0}{N}, \quad \lambda = \frac{\beta d}{L}$$

a: slit-to-biprism distance; b: biprism-to-observation distance; L: total source-to-screen distance; x_0, x_N: micrometer positions N fringe intervals apart; d: virtual-source separation; d_1, d_2: image separations at conjugate lens positions; u, v: positive object/image distance magnitudes.

Newton's rings

O: 42-46

Topic symbols. r, D: ring radius/diameter; t: local film thickness.

Dark & bright reflected rings

$$r^2_{m,\text{dark}} = \frac{m\lambda R}{n}, \quad D^2_{m,\text{dark}} = \frac{4m\lambda R}{n} \quad (m = 0, 1, \ldots)$$

$$r^2_{m,\text{bright}} = \frac{(m-1/2)\lambda R}{n}, \quad D^2_{m,\text{bright}} = \frac{4(m-1/2)\lambda R}{n} \quad (m \geq 1)$$

m: ring order; λ: vacuum wavelength; n: film index (1 for air); R: lens curvature radius; r_m, D_m: ring radius and diameter. Contact at centre; one reflection phase reversal; centre dark in reflection.

Difference method: either like-kind ring sequence

$$D^2_{m+p} - D^2_m = \frac{4p\lambda R}{n}, \quad r^2_{m+p} - r^2_m = \frac{p\lambda R}{n}$$

$$\lambda = \frac{n(D^2_{m+p} - D^2_m)}{4pR}, \quad R = \frac{n(D^2_{m+p} - D^2_m)}{4p\lambda}$$

p: positive integer difference in ring orders; m: initial order. Compare dark with dark or bright with bright; other symbols defined in this topic's formula groups.

Use: differences of squared ring sizes determine wavelength or curvature while reducing a constant central-gap offset.

Liquid index & dark ring order

$$n_{\text{liq}} \simeq \frac{D^2_{m,\text{air}}}{D^2_{m,\text{liq}}}, \quad m_{\text{dark}} = \frac{nD^2_m}{4\lambda R}$$

n_{liq}: liquid index (air index approximated as 1); $D_{m,\text{air}}, D_{m,\text{liq}}$: same-order diameters before/after filling. Absolute order formula assumes perfect central contact; a fractional dark order signals inconsistent data or a central gap.

Number of complete bright rings in an aperture

$$m_{\text{max}} = \left\lfloor \frac{nr^2_{\text{max}}}{\lambda R} + \frac{1}{2} \right\rfloor$$

r_{max}: available lens/contact-region radius; m_{max}: count of bright rings; $\lfloor \cdot \rfloor$: greatest integer not exceeding its argument. Ideal contact; normal incidence; ring centres counted only within the available radius.

Single-slit & circular diffraction

O: 43; 47-48; 51-54

Topic symbols. a: slit width; λ: wavelength; θ: diffraction angle; I_0: central intensity; $\alpha = \pi a \sin\theta/\lambda$

Slit minima & approximate bright positions

$$a \sin\theta_j = j\lambda \quad (j = \pm 1, \pm 2, \ldots)$$

$$\alpha_j \simeq (j + 1/2)\pi \quad (j = 1, 2, \ldots)$$

j: fringe order; α_j: exact nonzero root for a secondary maximum, or its midpoint approximation as marked. Exact first two positive roots: 4.4934, 7.7253; corresponding intensity ratios 0.0472, 0.0165.

Screen positions & central width

$$x_j \simeq \frac{j\lambda F}{a}, \quad W_{\text{central}} \simeq \frac{2\lambda F}{a}, \quad \Delta\theta_{\text{central}} \simeq \frac{2\lambda}{a}$$

$$\text{Separation of } \pm j \text{ minima} = \frac{2j\lambda F}{a}$$

F: focusing-lens focal length, or far-field screen distance when appropriate; x_j: minimum coordinate; W_{central}: full central-maximum width; $\Delta\theta_{\text{central}}$: angular full width. Small angles.

Airy-disc radius & diameter

$$\sin\theta_1 = 1.21967\frac{\lambda}{D}, \quad r_A \simeq 1.22\frac{F\lambda}{D}, \quad D_A \simeq 2.44\frac{F\lambda}{D}$$

D: circular aperture diameter; J_1: first-order Bessel function; z: dimensionless argument; θ_1: first dark-ring angle; F: focal length; r_A, D_A: Airy-disc radius and diameter. The bank also uses $F\lambda/D$ as a slit-like estimate; it omits the circular-aperture factor 1.22.

Meaning: the Airy disc sets the circular-aperture diffraction blur; its first-dark-ring factor is 1.22.

Double slit & diffraction grating

O: 48-53

Topic symbols. a, b: slit width/opaque gap; $d = a + b$: pitch; N: illuminated lines; λ: wavelength; $\alpha = \pi a \sin\theta/\lambda$; $\beta = \pi d \sin\theta/\lambda$.

Interference orders & missing orders

$$d\sin\theta = m\lambda \text{ (bright)}, \quad d\sin\theta = (m + 1/2)\lambda \text{ (dark)}$$

$$a\sin\theta = j\lambda \text{ (envelope zero)}, \quad m = \frac{d}{a}j \text{ (missing)}$$

m: integer interference order; j: nonzero integer diffraction-minimum order. A missing order requires both conditions simultaneously with integer orders. a, d, θ, λ defined in this topic's formula groups.

Meaning: an order disappears when an interference maximum lands on a single-slit diffraction minimum.

Pitch, line count & highest order

$$d = \frac{1}{\sigma}, \quad N = \frac{W}{d} = \sigma W, \quad \theta_m = \sin^{-1}\frac{m\lambda}{d}$$

$$|m|\lambda \le d, \quad m_{\max} = \left\lfloor \frac{d}{\lambda} \right\rfloor, \quad \mathcal{N}_{\text{directions}} = 2m_{\max} + 1$$

σ: line density (m^{-1}); W: illuminated grating width; d: pitch; m: spectral order; θ_m: direction; $\mathcal{N}_{\text{directions}}$: geometric order count before missing orders. Equality gives grazing emergence; practical observable count may be lower.

Exact angular separation of two wavelengths

$$\Delta\theta = \left| \sin^{-1}\frac{m\lambda_2}{d} - \sin^{-1}\frac{m\lambda_1}{d} \right|$$

m: fixed spectral order; $\Delta\lambda = \lambda_2 - \lambda_1$: wavelength difference; $\Delta\theta$: angular separation (rad); d: pitch; θ: mean diffraction angle for the small-separation approximation.

Resolution & polarization

O: 50-58

Topic symbols. $\Delta\lambda$: wavelength separation.

Telescope: Rayleigh criterion

$$\theta_R \simeq 1.22\frac{\lambda}{D}, \quad s_{\min} \simeq \ell\theta_R, \quad \Delta x \simeq F\theta_R, \quad \mathcal{R}_\theta = \frac{1}{\theta_R}$$

θ_R: angular resolution limit (rad); λ: wavelength; D: objective diameter; ℓ: object distance; $s_{\min}$: resolvable object separation; F: focal length; Δx: focal-plane separation; $\mathcal{R}_\theta$: angular resolving power.

Grating spectral resolution & required lines

$$\mathcal{R}_\lambda = \frac{\lambda}{\Delta\lambda} = |m|N, \quad N_{\min} = \left\lceil \frac{\lambda}{|m|\Delta\lambda} \right\rceil$$

n: object-space index; α: objective half-aperture angle; D, F: objective diameter/focal length; NA: numerical aperture; d: resolution distance. The bank uses the Abbe form; point-source Rayleigh uses 0.61. $\mathcal{R}_\lambda$: grating spectral resolving power; m: nonzero order; N: illuminated lines; $\lceil\cdot\rceil$: round up.

Meaning: resolving power quantifies the ability to distinguish nearby details or wavelengths; a smaller resolution limit is better.

Malus law & ideal polarizer chain

$$I = I_p\cos^2\theta, \quad I_p = \frac{I_u}{2} \text{ (unpolarized input)}$$

$$I_{\text{out}} = \frac{I_u}{2}\prod_{j=1}^{N-1}\cos^2\theta_j$$

I_p: incident linearly polarized intensity at analyzer; I_u: unpolarized input intensity; I: analyzer output; θ: angle between incident polarization and analyzer axis; N: number of ideal polarizers; θ_j: relative angle between adjacent axes.

Wave plates & optical activity

O: 55-58

Topic symbols. δ: retardation

Quarter-wave plate thickness & retardation

$$t_{\text{QWP}} = \frac{(2j+1)\lambda}{4\Delta n}, \quad \delta_{\text{QWP}} = (2j+1)\frac{\pi}{2} \qquad (j = 0, 1, \ldots)$$

j: plate order; t: thickness; Δn: birefringence magnitude; λ: wavelength. Minimum-thickness plates use $j = 0$. QWP at $45°$ to principal axes gives circular polarization; axis-aligned input remains linear.

Meaning: a wave plate delays one principal polarization component relative to the other, changing the polarization state.

Specific rotation & concentration

$$[\alpha]_\lambda^T = \frac{\theta}{\ell c_s}, \quad \theta = [\alpha]_\lambda^T \ell c_s, \quad c_s = \frac{\theta}{[\alpha]_\lambda^T \ell}$$

$$\frac{\theta_1}{\ell_1 c_1} = \frac{\theta_2}{\ell_2 c_2}, \quad c_2 = c_1 \frac{\theta_2 \ell_1}{\theta_1 \ell_2}$$

$[\alpha]_\lambda^T$: specific rotation at wavelength λ and temperature T; θ: observed rotation (degrees); ℓ: solution path length (dm); c_s: solute concentration (g mL^{-1}). Specific rotation unit: degree mL g^{-1} dm^{-1}. Subscripts distinguish two measurements of the same substance at the same T, λ.

Use: separates the intrinsic optical rotation of a solution from the effects of concentration and path length.

Kinetic theory & molecular speeds

T: 4-12

Topic symbols. c: molecular speed.

Ideal gas, density & molecular count

$$PV = nRT = Nk_BT, \quad N = nN_A, \quad R = N_A k_B$$

$$\rho = \frac{Nm}{V} = \frac{nM}{V}, \quad n_N = \frac{N}{V} = \frac{P}{k_B T}$$

P: pressure (Pa); V: volume (m^3); T: absolute temperature (K); n: amount (mol); N: molecule count; N_A: Avogadro constant; R: universal gas constant; k_B: Boltzmann constant; m: mass per molecule (kg); M: molar mass (kg mol^{-1}); ρ: density; n_N: number density (m^{-3}).

Pressure from molecular motion

$$P = \frac{1}{3}\rho\overline{c^2} = \frac{Nm\overline{c^2}}{3V}, \qquad c_{\text{rms}} = \sqrt{\frac{3P}{\rho}}$$

Δp_x: momentum transferred per elastic collision; Δt: interval between hits on the same wall; ℓ: cube side; F_i: mean force from one molecule. m: molecular mass; other symbols defined in this topic's formula groups.

Translational energy & temperature

$$\bar{\varepsilon}_{\text{tr}} = \frac{1}{2}m\overline{c^2} = \frac{3}{2}k_BT, \quad U_{\text{tr}} = \frac{3}{2}nRT = \frac{3}{2}PV$$

$$c_{\text{rms}} = \sqrt{\frac{3k_BT}{m}} = \sqrt{\frac{3RT}{M}}, \quad \frac{T_1}{M_1} = \frac{T_2}{M_2} \quad \text{(equal rms speed)}$$

$\bar{\varepsilon}_{\text{tr}}$: mean translational energy per molecule; U_{tr}: total translational energy. Subscripts 1,2 denote two gases; all temperatures absolute and molar masses in consistent units.

Molecular cross section & mean free path

$$\sigma_c = \pi d^2, \quad \lambda_{\text{mfp}} = \frac{1}{\sqrt{2}\,\pi d^2 n_N} = \frac{k_BT}{\sqrt{2}\,\pi d^2 P}$$

d: hard-sphere molecular diameter; σ_c: collision cross section; λ_{mfp}: mean free path; n_N: number density; $\bar{c}_{\text{rel}}$: mean relative speed of identical molecules; z_c: collision rate per molecule; τ_c: mean collision time. Stationary-target approximation omits $\sqrt{2}$.

Meaning: mean free path is the mean distance travelled between collisions, not the distance between neighbouring molecules.

Maxwell distributions

T: 7-12

Topic symbols. c, c_x, c_y, c_z: speed/velocity components; m: molecular mass; k_B: Boltzmann constant; T: temperature; N: molecule count

Most probable & rms speed

$$c_{\text{mp}} = \sqrt{\frac{2k_BT}{m}}, \qquad c_{\text{rms}} = \sqrt{\frac{3k_BT}{m}}$$

c_{mp}: speed at the peak of $f(c)$; $\bar{c}$: mean speed; c_{rms}: rms speed. Replace k_B/m by R/M for molar forms; R: gas constant; M: molar mass.

Meaning: most probable speed is the distribution peak; mean speed is the ordinary average; rms speed is the energy-related average.

Equipartition & heat capacities

T: 13-16

Equipartition, internal energy & molar capacities

$$\bar{\varepsilon} = \frac{f_E}{2}k_BT, \quad U = \frac{f_E}{2}nRT$$

$$C_{V,m} = \frac{f_E}{2}R, \quad C_{P,m} = \left(\frac{f_E}{2}+1\right)R, \quad \gamma = \frac{C_{P,m}}{C_{V,m}} = 1+\frac{2}{f_E}$$

$\bar{\varepsilon}$: mean molecular energy; U: gas internal energy; n: amount (mol); $C_{V,m}, C_{P,m}$: molar heat capacities (J mol^{-1} K^{-1}); γ: heat-capacity ratio; k_B, R, T: Boltzmann constant, gas constant, temperature. Ideal gas, classical active modes; fixed f_E over the range.

Meaning: equipartition assigns half kT to each active quadratic energy term; active vibrational modes contribute twice.

Common ideal-gas values

model	f_E	$C_{V,m}/R$	$C_{P,m}/R$	γ
monatomic	3	3/2	5/2	5/3
linear, vibration frozen	5	5/2	7/2	7/5
nonlinear, vibration frozen	6	3	4	4/3
diatomic, one active vibration	7	7/2	9/2	9/7

f_E: active quadratic terms; $C_{V,m}, C_{P,m}$: molar capacities; R: gas constant. Activation temperatures depend on the particular molecule; these are model limits.

Mixture heat capacities & gamma

$$C_{V,\text{tot}} = \sum_i n_i C_{V,m,i}, \quad C_{P,\text{tot}} = \sum_i n_i C_{P,m,i}, \quad \gamma_{\text{mix}} = \frac{\sum_i n_i C_{P,m,i}}{\sum_i n_i C_{V,m,i}}$$

$$\gamma_{\text{mono+di}} = \frac{5n_1+7n_2}{3n_1+5n_2}$$

n_i: moles of species i; $C_{V,m,i}, C_{P,m,i}$: species molar capacities; $C_{V,\text{tot}}, C_{P,\text{tot}}$: total mixture capacities (J K^{-1}). Last formula: n_1 monatomic, n_2 diatomic, vibrations frozen.

Real gases & critical constants

T: 16-20; 48

Topic symbols. P, V, T: pressure, total volume, temperature; n: moles; $v_m = V/n$: molar volume; a, b: van der Waals parameters; R: gas constant

Critical temperature from pressure & density

$$\frac{RT_c}{P_cv_c} = \frac{8}{3}, \quad v_c = \frac{M}{\rho_c}, \quad T_c = \frac{8P_cM}{3R\rho_c}$$

P_c, T_c, v_c: critical model values; Z_c: critical compressibility factor; M: molar mass; ρ_c: critical density. These van der Waals predictions need not match experimental critical values exactly.

Meaning: the critical constants locate the model point where liquid and vapour cease to be distinct phases.

First law & thermodynamic processes

T: 20-28

Topic symbols. W: work by gas

First law & boundary work

$$\Delta U = Q - W_{\text{by}} = Q + W_{\text{on}}, \quad W_{\text{on}} = -W_{\text{by}}$$

$$\delta W_{\text{by}} = P_{\text{ext}}\,dV, \quad W_{\text{by}} = \int_{V_1}^{V_2} P_{\text{ext}}\,dV, \quad dU = \delta Q - P\,dV \text{ (reversible)}$$

U: internal energy; Q: heat into system; W_{by}: work by system; W_{on}: work on system (all J); P_{ext}: external pressure; P: system pressure; V: volume. δ denotes a path-dependent increment.

Capacity conventions & Mayer relation

$$C_{V,\text{tot}} = \left(\frac{\partial U}{\partial T}\right)_V = nC_{V,m} = m_g c_v$$

$$C_{P,\text{tot}} = \left(\frac{\partial H}{\partial T}\right)_P = nC_{P,m} = m_g c_p$$

$$C_{P,m} - C_{V,m} = R, \quad c_p - c_v = R_s = \frac{R}{M} = \frac{P}{\rho T}$$

H: enthalpy; T: temperature; $C_{V,\text{tot}}, C_{P,\text{tot}}$: total capacities (J K^{-1}); $C_{V,m}, C_{P,m}$: molar capacities; m_g: gas mass; c_v, c_p: mass-specific capacities; R_s: specific gas constant; M: molar mass; ρ: density. Mayer relations here are ideal-gas relations.

Meaning: heat capacity measures heat needed for a unit temperature rise under the stated constraint.

Ideal gas: energy & isochoric / isobaric heat

$$\Delta U = nC_{V,m}(T_2 - T_1), \quad \Delta H = nC_{P,m}(T_2 - T_1)$$

$$V = \text{const}: \quad W = 0, \quad Q = \Delta U, \quad \frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$P = \text{const}: \quad W = P(V_2 - V_1) = nR(T_2 - T_1), \quad Q = \Delta H$$

$W = W_{\text{by}}$; subscripts 1,2: initial/final states; n: moles; R: gas constant; P, V, T, U, H: pressure, volume, temperature, energy, enthalpy. Capacities constant over the interval.

Isothermal ideal gas

$$T = \text{const}: \quad P_1V_1 = P_2V_2, \quad \Delta U = 0$$

$$W_{\text{by}} = Q = nRT \ln\frac{V_2}{V_1} = nRT \ln\frac{P_1}{P_2}$$

n: amount; R: gas constant; T: common temperature; P_i, V_i: endpoint states; Q: heat in. Work formula assumes a reversible or quasistatic mechanically balanced isothermal path.

Cycle work, straight-line work & rectangle area

$$\oint dU = 0, \quad Q_{\text{net}} = W_{\text{net}}, \quad W_{\text{cycle}} = \oint P\,dV$$

$$W_{\text{straight}} = \frac{P_1 + P_2}{2}(V_2 - V_1), \quad W_{\text{rectangle}} = (P_H - P_L)(V_H - V_L)$$

$Q_{\text{net}}, W_{\text{net}}$: signed cycle totals; P_H, P_L: high/low rectangle pressures; V_H, V_L: large/small volumes. Rectangle formula positive for clockwise traversal. Ideal-gas free expansion also gives $\Delta T = 0$.

Adiabatic processes & steady flow

T: 23-28

Topic symbols. n: moles; R: gas constant; γ: capacity ratio; Q: heat in; W: work by gas; h: specific enthalpy; w: flow speed.

Reversible adiabatic ideal gas

$$PV^{\gamma} = \text{const}, \quad TV^{\gamma-1} = \text{const}, \quad T^{\gamma}P^{1-\gamma} = \text{const}$$

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} = \left(\frac{P_2}{P_1}\right)^{(\gamma-1)/\gamma}$$

P, V, T: pressure, volume, temperature; $\gamma = C_{P,m}/C_{V,m}$: heat-capacity ratio; subscripts 1,2: endpoint states. Ideal gas, constant heat capacities, reversible adiabatic path. Sudden bursts in the bank use this as an idealized model; no-heat exchange alone does not guarantee these relations.

Adiabatic horizontal nozzle

$$q = w_s = 0, \quad z_1 = z_2: \quad h_1 + \frac{w_1^2}{2} = h_2 + \frac{w_2^2}{2}$$

$$w_2 = \sqrt{w_1^2 + 2(h_1 - h_2)}, \qquad w_1 \simeq 0: \quad w_2 \simeq \sqrt{2(h_1 - h_2)}$$

h_1, h_2: inlet/outlet specific enthalpies in J kg^{-1}; w_1, w_2: speeds in m s^{-1}. Multiply enthalpy in kJ kg^{-1} by 10^3 before using these speed formulas.
Use: converts a nozzle's enthalpy drop into an increase in flow kinetic energy.

Heat engines & Carnot cycle

T: 29-37

Topic symbols. T_H, T_L: hot/cold temperatures; n: moles; R: gas constant

General heat engine & Carnot bound

$$W = Q_H - Q_L, \quad \eta = \frac{W}{Q_H} = 1 - \frac{Q_L}{Q_H}$$

$$\eta \le \eta_C = 1 - \frac{T_L}{T_H}, \quad \frac{Q_H}{T_H} = \frac{Q_L}{T_L} \quad \text{(reversible)}$$

$Q_H, Q_L > 0$: absorbed and rejected heat magnitudes per cycle; W: net work output; η: efficiency; $T_H > T_L$: reservoir temperatures (K); η_C: Carnot efficiency. Equality for a reversible engine.
Meaning: efficiency is the fraction of absorbed heat converted to net cycle work. Carnot efficiency is the reversible limit for two reservoirs.

Reservoir temperatures & useful rearrangements

$$T_H = \frac{T_L}{1 - \eta_C}, \quad T_L = (1 - \eta_C)T_H$$

$$\Delta T = T_H - T_L: \quad T_H = \frac{\Delta T}{\eta_C}, \quad T_L = \frac{(1 - \eta_C)\Delta T}{\eta_C}$$

$$Q_L = Q_H \frac{T_L}{T_H}, \quad W = \eta_C Q_H$$

ΔT: reservoir temperature separation; all other symbols defined in this topic's formula groups. Use efficiency as a fraction, not a percentage number.

Source-temperature change at a fixed sink

$$T_H' - T_H = T_L \left(\frac{1}{1 - \eta_C'} - \frac{1}{1 - \eta_C}\right)$$

$\delta T > 0$: equal small source increase or sink decrease; prime: new condition; η_C: Carnot efficiency. Source-change expression with $\simeq$ is a small-change approximation; fixed-T_H sink-change expression is exact.

Two isobars joined by reversible adiabatics

$$\eta = 1 - \left(\frac{P_L}{P_H}\right)^{(\gamma-1)/\gamma}, \quad \gamma = \frac{\ln(P_B/P_C)}{\ln(V_C/V_B)}$$

P_H, P_L: high and low cycle pressures; γ: constant ideal-gas heat-capacity ratio. $B \to C$: adiabatic expansion between the isobars; V_B, V_C: volumes. Applies to the bank cycle with isobaric heat addition/rejection.

Refrigerators & entropy laws

T: 34-40

Topic symbols. $Q_H, Q_L > 0$: heat magnitudes; S: entropy; S_{gen}: entropy generation; T_b: boundary temperature; COP: heat moved per work input.

Refrigerator work & coefficient of performance

$$W_{\text{in}} = Q_H - Q_L, \quad \text{COP}_R = \frac{Q_L}{W_{\text{in}}}, \quad \text{COP}_{R,C} = \frac{T_L}{T_H - T_L}$$

$Q_L > 0$: heat extracted from cold body; $Q_H > 0$: heat delivered to hot body; W_{in}: input work; COP: coefficient of performance; subscripts R, HP, C: refrigerator, heat pump, reversible Carnot limit. T_H, T_L: temperatures in K. Meaning: refrigerator COP is heat removed from the cold space per unit input work; it can exceed one.

Entropy changes & phase transitions

T: 40-43

Topic symbols. L: specific latent heat.

Ideal gas: entropy in temperature-pressure or pressure-volume

$$\Delta S = nC_{P,m} \ln \frac{T_2}{T_1} - nR \ln \frac{P_2}{P_1}$$
$$= nC_{V,m} \ln \frac{P_2}{P_1} + nC_{P,m} \ln \frac{V_2}{V_1}$$

ΔS: total entropy change; n: moles; $C_{V,m}, C_{P,m}$: constant molar capacities; R: gas constant; P_i, V_i, T_i: endpoint states. Applies to any path between equilibrium ideal-gas states; use a reversible path for evaluating entropy.

Variable heat capacity: temperature-pressure form

$$\Delta S = n \left[a \ln \frac{T_2}{T_1} + \kappa(T_2 - T_1) - R \ln \frac{P_2}{P_1} \right]$$

a, b: molar-capacity intercepts (J mol^{-1} K^{-1}); κ: common slope (J mol^{-1} K^{-2}). For mass-specific versions replace $C_{P,m}, C_{V,m}, R, n, \Delta S$ by $c_p, c_v, R_s, 1, \Delta s$ consistently.

Phase change at fixed transition temperature

$$Q = m_g L, \quad \Delta S = \frac{m_g L}{T_{\text{tr}}}, \qquad \Delta S_{\text{condensation/freezing}} = -\frac{m_g L}{T_{\text{tr}}}$$

m_g: mass transformed; $L > 0$: latent heat per mass (J kg^{-1}); T_{tr}: transition temperature (K). Positive sign for fusion/vaporization; negative for reverse change; equilibrium transition.

Ice to steam: sum of heating & latent contributions

$$\Delta S = m_g \left[c_i \ln \frac{T_m}{T_i} + \frac{L_f}{T_m} + c_w \ln \frac{T_b}{T_m} + \frac{L_v}{T_b} \right]$$

T_i: initial ice temperature; T_m: melting temperature; T_b: boiling temperature; c_i, c_w: constant specific heats of ice/water; L_f, L_v: latent heats of fusion/vaporization; m_g: mass. Final steam at T_b; all temperatures in K, compatible heat units.

Phase equilibrium & Clapeyron equation

T: 49-54

Topic symbols. T, P: coexistence temperature/pressure; R: gas constant

Clapeyron coexistence slope

$$\frac{dP_{\text{sat}}}{dT} = \frac{s_2 - s_1}{v_2 - v_1} = \frac{L}{T(v_2 - v_1)}$$

g_1, g_2: specific Gibbs free energies (J kg^{-1}); s_1, s_2: specific entropies (J kg^{-1} K^{-1}); v_1, v_2: specific volumes (m^3 kg^{-1}); L: latent heat from phase 1 to 2 (J kg^{-1}); P_{sat}: coexistence pressure; T: transition temperature.
Meaning: the Clapeyron slope tells how the equilibrium melting or boiling temperature changes with pressure.

Small transition-temperature / pressure shifts

$$\Delta T \simeq \frac{T(v_2 - v_1)}{L}\Delta P, \qquad \Delta P \simeq \frac{L}{T(v_2 - v_1)}\Delta T$$

$$P_{\text{final}} = P_{\text{initial}} + \Delta P$$

$\Delta T, \Delta P$: small coexistence shifts; T, L, v_1, v_2: values near the initial state. For ice melting $v_2 - v_1 < 0$; for vaporization $v_2 - v_1 > 0$. Pressure increase and final absolute pressure are distinct.

Integrated Clausius-Clapeyron plot

$$\ln \frac{P_2}{P_1} = -\frac{L_m}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

$$\ln \frac{P}{P_*} = -\frac{L_m}{R}\frac{1}{T} + C, \quad L_m = -R\frac{d\ln(P/P_*)}{d(1/T)}$$

$$L_m = -2.303R\frac{d\log_{10}(P/P_*)}{d(1/T)}$$

P_1, P_2, T_1, T_2: two coexistence states; P_*: fixed reference pressure (same units as P); C: dimensionless integration constant; L_m: latent heat assumed constant over fitted range. Plot axes matter: for horizontal $10^3/T$, multiply the fitted slope by 10^3 before using $-R\times$slope.

Use: a straight-line vapour-pressure plot yields latent heat from its slope.

Latent-heat basis & heat-unit conversion

$$L_m = ML, \quad \frac{dP}{dT} = \frac{JL_{\text{heat}}}{T(v_2 - v_1)}$$

M: molar mass (kg mol^{-1}); L: latent heat (J kg^{-1}); L_{heat}: latent heat in calorie units per compatible mass; J: mechanical equivalent (4.184 J cal^{-1}, or the rounded value supplied by the exam). Do not multiply by J if latent heat is already in joules.

Constants, units & notation

O & T: numerical sections throughout

Topic symbols. Use the unit system and rounded constants stated in each exam question. All thermodynamic temperatures are absolute unless explicitly labelled Celsius.

Physical constants

$$k_B = 1.380649 \times 10^{-23}\ \text{J K}^{-1}, \quad N_A = 6.02214076 \times 10^{23}\ \text{mol}^{-1}$$

$$R = 8.314462618\ldots\ \text{J mol}^{-1}\text{K}^{-1}, \quad h = 6.62607015 \times 10^{-34}\ \text{J s}$$

$$c = 299\,792\,458\ \text{m s}^{-1}, \quad e = 1.602176634 \times 10^{-19}\ \text{C}$$

k_B: Boltzmann constant; N_A: Avogadro constant; $R = N_A k_B$: gas constant; h: Planck constant; c: vacuum light speed; e: elementary charge magnitude. Use question-specified rounded values for numerical exam work.

Pressure, temperature & hydrostatics

$$T\,[\text{K}] = t_C\,[^\circ\text{C}] + 273.15, \quad \Delta P = \rho g h$$

$$1\ \text{atm} = 101325\ \text{Pa}, \quad 1\ \text{bar} = 10^5\ \text{Pa}$$

$$1\ \text{Torr} = \frac{101325}{760}\ \text{Pa} \simeq 133.322\ \text{Pa}$$

T: absolute temperature; t_C: Celsius reading; ρ: manometer-fluid density; g: gravity (use 9.8 or 9.81 m s^{-2} as specified); h: column height difference. Gas relations and efficiency use absolute pressures.

PART II / FROM NUMERICAL PROBLEMS

Length, volume, energy & concentration

$$1\ \text{Å} = 10^{-10}\ \text{m}, \quad 1\ \text{nm} = 10^{-9}\ \text{m}, \quad 1\ \mu\text{m} = 10^{-6}\ \text{m}$$

$$1\ \text{L} = 10^{-3}\ \text{m}^3, \quad 1\ \text{cm}^3 = 10^{-6}\ \text{m}^3, \quad 1\ \text{erg} = 10^{-7}\ \text{J}$$

$$1\ \text{cal} = 4.184\ \text{J}, \quad 1\ \text{g cm}^{-3} = 10^{3}\ \text{kg m}^{-3}$$

$$1\ \text{g mL}^{-1} = 10^{3}\ \text{g L}^{-1}, \qquad c_{\%(w/v)} = 100 c_{\text{g/mL}}$$

Angstrom ($\mathring{A}$), nanometre (nm), micrometre (μm); litre (L); erg and calorie (cal): energy units. $c_{\%(w/v)}$: grams solute per 100 mL solution. Use the calorie convention supplied in the question when it differs.

Notation conventions used across this sheet

$$\omega = 2\pi f, \quad k_w = \frac{2\pi}{\lambda}, \quad \hbar = \frac{h}{2\pi}, \quad \ln x = 2.302585\ldots \log_{10} x$$

f: frequency; ω: angular frequency; k_w: wavenumber (distinct from spring stiffness k); λ: wavelength except λ_{mfp}; γ_d: damping rate (distinct from thermal γ); ω_e: optical dispersive power; ln: natural logarithm. $\simeq$ marks approximation; derivatives hold subscripted variables fixed.